Particle Physics 2 Standard Model

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Chapter 1

Lecture 1: New Revolutions in Particle Physics: Standard Model

These notes follow Leonard Susskind's first lecture in *Particle Physics 2: Standard Model*, with curation by LazyingArt LLC. The lecture begins by delaying, once again, the promised particle list. That delay is not a digression. We already have two sides of the modern picture, particles and fields; before the catalogue can begin in earnest, we need the third side as well, namely force. The lecture therefore unfolds in a very definite order: first the field-language account of force, then the exchange-language account, and only then the first entries of the particle table.

1.1 Particles, Fields, and the Missing Third Leg

By this point in the course we have already built a good deal of machinery. We have fields and their quanta, wave equations and action principles, and the broad kinematic properties of particles: energy, momentum, spin, and the boson-fermion distinction. Fermions resist occupancy of the same state; bosons do not, and may pile up coherently into one state. We also know that products of fields in a Lagrangian encode processes, or vertices, in which particles come in and particles go out.

But something is still missing. Before we start naming the particles one by one, or rather group by group, the lecture insists that we complete a triangle of ideas:

- particles,
- fields for which those particles are quanta,
- forces associated with exchange processes involving those same quanta.

There is also a warning here, and it is worth preserving. The distinction between elementary and composite particles is not settled at the outset. We postpone it. For now we imagine that there are fundamental fields and that their quanta are elementary particles. The missing issue is different: if every field corresponds to a particle, and if every familiar field gives rise to a force, then there must also be a way to think about force directly in particle language.

That is the theme of the lecture. We are not going to be satisfied with Coulomb's law as a memorized formula. We want to see where a force comes from in the field description, and then see how the same phenomenon is retold when the field is regarded as a collection of quanta.

1.2 Force from Field Energy in Classical Electrodynamics

Susskind first closes the triangle in the classical language. Suppose we place two charges in space and ask for the force between them. One answer is to quote the inverse-square law. The lecture chooses a better answer: ask what the charges do to space.

They create electric fields. If the first charge produces $\mathbf{E}_1$ and the second produces $\mathbf{E}_2$, then the total field is

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2. \tag{1.1}$$

The transcript here is garbled in the details of a temporary charge-rescaled notation, so we keep only the stable mathematical core. The field energy is obtained schematically by squaring the electric field and integrating over space:

$$U_{\text{field}} \propto \int d^3x\, \mathbf{E}^2. \tag{1.2}$$

Substituting the sum of the two fields gives

$$U_{\text{field}} \propto \int d^3x\, (\mathbf{E}_1 + \mathbf{E}_2)^2 \tag{1.3}$$

$$\propto \int d^3x\, (\mathbf{E}_1^2 + \mathbf{E}_2^2 + 2\,\mathbf{E}_1 \cdot \mathbf{E}_2). \tag{1.4}$$

Now the lecture pauses and reads this expansion term by term. The first term is the self-energy of particle 1. The second is the self-energy of particle 2. Each is already part of the energy, hence the mass, of the corresponding isolated particle. Each is position-independent. For the present purpose, we set them aside.

The cross-term is different. It is not the energy of either particle separately. It exists only because both are present. If the charges are extremely far apart, the two Coulomb fields scarcely overlap and the integral is negligible. If they are brought closer, the overlap grows, and so does the new contribution. Thus the natural interaction energy is

$$U_{\text{int}}(r) \propto \int d^3x\, \mathbf{E}_1 \cdot \mathbf{E}_2. \tag{1.5}$$

The force is then read from the r-dependence of this interaction energy:

$$F(r) = -\frac{dU_{\text{int}}}{dr}. \tag{1.6}$$

If one performs the full calculation, one recovers the usual Coulomb behavior,

$$F_{\text{Coulomb}}(r) \propto \frac{q_1 q_2}{r^2}. \tag{1.7}$$

This is the lecture's first important payoff. The force law is now understood as coming from the portion of the field energy that depends on the relative arrangement of the two charges. One could say it this way: bring the charges together, the field in between them is distorted, and that distortion carries energy. The force is the derivative of that energy.

But if fields are nothing but collections of quanta, then this cannot be the end of the story. There must also be a particle-language account of the same force.

1.3 Molecular Exchange: Two Protons and One Electron

Before returning to electrodynamics in that second language, the lecture takes a detour to a system where the exchange picture is easier to see. Consider two protons and one electron. This is not chosen because it is the same as electrodynamics. It is chosen because the quantum mechanics is more intuitive.

If one proton is present and the other is taken very far away, the electron occupies the hydrogenic ground state centered on the first proton. The board first records this simply as a localized hump.

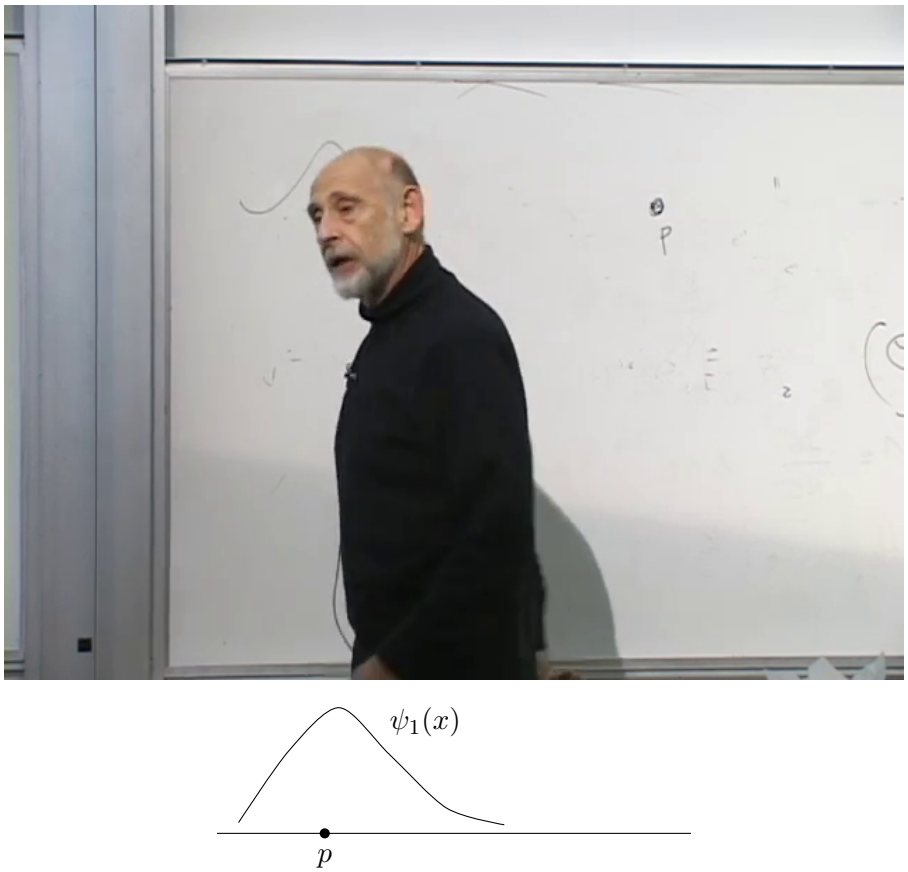
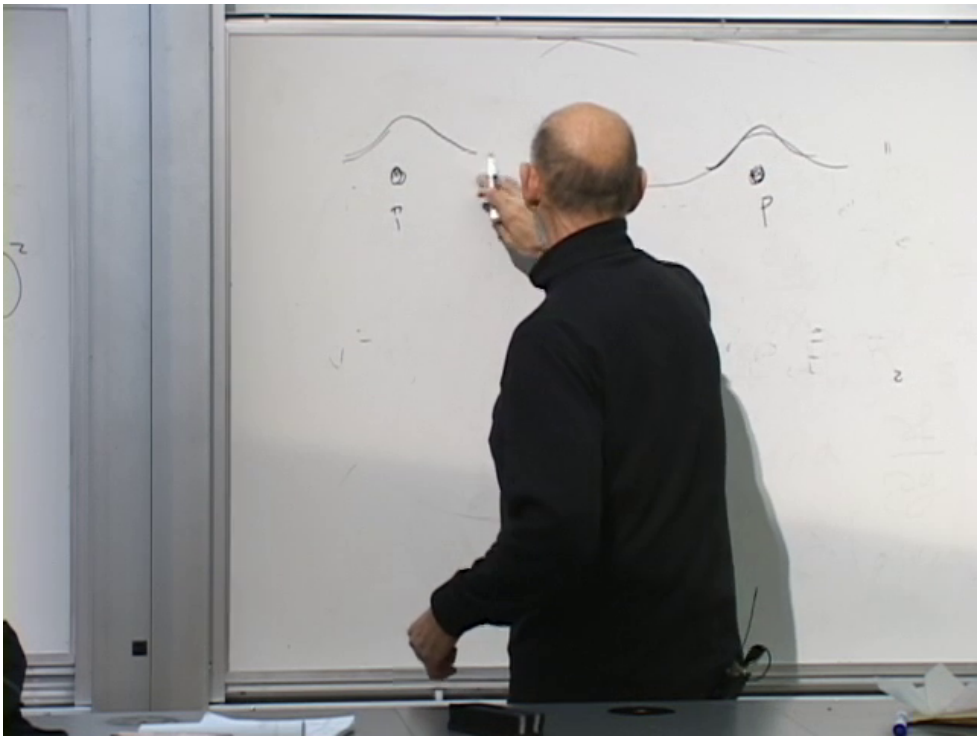


Figure 1.1: Single hydrogenic wave-function sketch. The frame documents the board layout; the TikZ curve is a cleaned schematic reading rather than a literal transcription of notation.

The faint point label beneath the chalk mark looks like a handwritten P, but it is too uncertain to be used as reliable notation. What matters is the shape: a bound-state profile localized near one force center.

Now place a second proton at finite but large separation. By symmetry there is a second localized state of the same energy, centered near the right proton.

At infinite separation the two states are degenerate. The electron may be localized near the left proton or near the right proton, and the energy is the same.



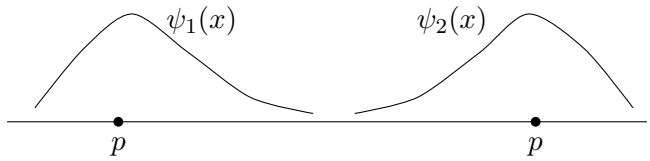


Figure 1.2: Two localized electron wave functions. The lecture now moves from these two one-center states to an equilibrium superposition.

1.3.1 Question & Answer

Question. Why is the two-proton system not exactly degenerate once the separation is finite?

Answer. Because once the protons are not infinitely far apart, processes become possible that were impossible before. In particular, the electron can tunnel from one center to the other. The lecture first describes this qualitatively, but the minimal mathematical reconstruction is a two-state Hamiltonian. If $|L\rangle$ and $|R\rangle$ denote the left- and right-localized states, then finite separation introduces off-diagonal matrix elements:

$$H|L\rangle = E_0|L\rangle - t|R\rangle, \tag{1.8}$$
$$H|R\rangle = E_0|R\rangle - t|L\rangle, \tag{1.9}$$

with $t > 0$ a tunneling amplitude.

The eigenstates are therefore the symmetric and antisymmetric combinations

$$|+\rangle = \frac{|L\rangle + |R\rangle}{\sqrt{2}}, \qquad E_+ = E_0 - t, \tag{1.10}$$

$$|-\rangle = \frac{|L\rangle - |R\rangle}{\sqrt{2}}, \qquad E_- = E_0 + t. \tag{1.11}$$

Translated back into wave functions, the lower equilibrium state is approximately

$$\psi_+(x) \approx \frac{\psi_1(x) + \psi_2(x)}{\sqrt{2}}. \tag{1.12}$$

The word "approximately" matters. The lecture is explicit that the true equilibrium wave function is not *exactly* the naive sum. In the overlap region between the two wells it is slightly different. The accepted frame does not settle the exact contour there, and in any case Susskind himself says he did not draw it very well. But the physical point is stable: once tunneling is allowed, the true equilibrium state has lower energy than the two isolated one-center states.

The lecture also says what one should not do. One should not picture this as a literal observed trajectory, with the electron watched in transit. As in the rest of quantum mechanics, trying to watch the tunneling ruins the phenomenon. The exchange picture is a way of describing the state, not a classical movie of a particle moving through space.

1.4 From Lowered Energy to an Effective Force

Now the lecture shifts, very deliberately, from the shape of the state to the shape of the energy. At large proton separation the electron energy is just the familiar one-center ground-state energy. At smaller separation, once tunneling and superposition are allowed, the equilibrium energy is slightly lower. Therefore the energy of the whole two-proton one-electron system becomes a function of the proton separation r.

This is the exact point at which the molecular example reconnects with the field-theory example. We may regard that separation-dependent energy as an effective potential $V(r)$. The board then adds the second, lower sketch: a qualitative graph of energy against separation.

The force is now read from the energy gradient:

$$F(r) = -\frac{dV}{dr}. \tag{1.13}$$

The lecture phrases it in plain language: force pushes toward decreased energy. If the energy drops as the protons move together, then the system experiences an effective attraction.

This is the molecular version of the same logic we saw earlier for classical fields. In one case the energy change comes from the overlap of two electric fields. In the other it comes from the lowering of the quantum-mechanical equilibrium energy in a two-center problem. In both cases, force comes from a configuration-dependent energy.

Several audience questions sharpen the intended picture, and they are worth preserving because otherwise the discussion becomes too loose.

What the picture is not.

- It is not mainly a screening story. The essential point is not that the electron merely reduces the proton charge seen by the other proton.
- It is not a third independent force in addition to the familiar ones. Rather, the Schrödinger problem in the presence of two force centers has a lower equilibrium energy than the sum of two isolated one-center problems.
- It is not best described as ordinary two-particle entanglement. The lecture says more pointedly that the relevant structure is the presence of off-diagonal elements in the Hamiltonian.
- It is not a literal watched trajectory. The worldline picture is schematic, useful for intuition and dangerous if taken too seriously.

The lecture also emphasizes a limitation. In the one-electron system the two protons still carry net positive charge, so Coulomb repulsion remains. The exchange attraction is short-ranged; the electrostatic repulsion is longer-ranged. Thus attraction does not automatically mean stable binding at all separations. In a neutral two-electron system, by contrast, the long-range Coulomb piece is much less dominant, and the same exchange mechanism shows up more cleanly as ordinary molecular binding.

The exchange picture can nevertheless be drawn, and the lecture does draw it in words.

1.5 Photon Exchange and the First Entries of the Table

Now that the exchange picture has been made visible in a molecular setting, the lecture returns to electrodynamics. Can the Coulomb force be told in the same language? Yes. A charged particle interacts with the electromagnetic field, and the quanta of that field are photons. Classically one solves for an electric field around the charge. Quantum mechanically one thinks instead in terms of emission and absorption of photons.

The lecture makes this more concrete. A physical charged particle is not just a bare charge. It is a quantum-mechanical superposition of states containing no photons, one emitted photon, two photons, and so on, with photons continually emitted and reabsorbed. If another charged particle is nearby, then a photon emitted by one may be absorbed by the other. That shared-photon process changes the energy of the two-particle system away from the simple sum of two isolated energies. That energy shift is the Coulomb interaction in particle language.

This is the place where Susskind explicitly closes the triangle. Particles, fields, and forces are now in one-to-one-to-one correspondence. When people say there are only four forces in nature, the lecture pushes back. In the exchange sense, every exchangeable quantum can mediate a force in some appropriate context. Electrons do so in molecular physics. Photons do so in electrodynamics.

Only after this bridge has been built does the lecture finally allow itself to start the table of particles. And even here the style matters. He does not want to dump the whole Standard Model on the board at once. Better to proceed in small groups, because the full thing is, as he says, a monstrosity.

The accepted frame should remain visible, because it captures exactly that switch in board logic.

The board evidence supports the secure identification

$$\gamma \leftrightarrow A. \tag{1.14}$$

The lecture then explains in words that A is the electromagnetic vector potential. In a later, more covariant notation one would often write A_μ, but the board itself only visibly shows A, so that is the notation we keep at first pass.

The table columns are introduced in a specific order: name, symbol, type, charge, baryon number, and mass. Here "type" means simply boson or fermion. The first secure row is the photon row; the next row, for the electron and positron, comes from the transcript rather than from the crop itself.

Name	Symbol	Field	Type	Q	B	Mass
Photon	γ	A	boson	0	0	0
Electron/positron	$e^\pm$	ψ_e	fermion	∓ 1	0	0.51 MeV

Table 1.1: A cleaned continuation of the lecture's first table entries. Only the photon row is directly visible in the accepted crop; the electron row is added from the transcript.

Here the lecture slows down again for units and conventions. Electric charge is measured in units of the electron charge. Thus we simply write

$$Q(e^-) = -1, \qquad Q(e^+) = +1. \tag{1.15}$$

Likewise mass is measured in energy units, since $E = mc^2$. The historically natural unit is the electron volt, and in particle physics we quickly move to MeV and GeV. The electron mass is

$$m_e \approx 0.51\,\text{MeV}. \tag{1.16}$$

The lecture also remarks that it is conventional to speak of the electron as the particle and the positron as its antiparticle, though the relation is actually mutual; choosing which one to call "the particle" is a convention of language.

The elementary QED process is then introduced: one electron in, one electron out, one photon attached. By rearranging the legs of that one vertex one reads the same diagram as emission, absorption, or positron emission. At the lecture's level the interaction term is described schematically as a product of one electron field in, one electron field out, and one photon field:

$$\mathcal{L}_{\text{int}} \sim e\,\psi_e^\dagger \psi_e A. \tag{1.17}$$

This is intentionally schematic. A more modern covariant form would refine both the spinor structure and the photon field notation, but the lecture's real point is simpler:

$$\mathcal{A}(e \to e + \gamma) \propto e, \qquad P(e \to e + \gamma) \propto e^2. \tag{1.18}$$

A schematic vertex is therefore useful.

And the force between charges is the many-body consequence of such elementary emission and absorption events:

At this point the opening promise can finally be kept. We are ready to begin naming the players.

1.6 Quarks, Baryon Number, and the Puzzle of Families

The next move is historically important. In an earlier era one might have gone directly from electrons and photons to protons and neutrons. The lecture instead says: no, modern particle physics places quarks first. Protons and neutrons are to be understood, at least naively, as composites of quarks.

There are six quark flavors,

$$u,\ d,\ s,\ c,\ b,\ t, \tag{1.19}$$

and the lecture is almost amused by the naming conventions. Up and down are not really up and down. Strange is not stranger than the others. Charm, bottom, and top are historical accidents. The names are not the structure.

The structure comes in two parts. First, all quarks are fermions. Second, their charges repeat in pairs:

$$Q(u) = Q(c) = Q(t) = +\frac{2}{3}, \qquad Q(d) = Q(s) = Q(b) = -\frac{1}{3}. \tag{1.20}$$

Thus the trio (d, u), (s, c), and (b, t) exhibits a pattern of replicated charge assignments. But the masses do not remotely repeat:

$$m_u \sim 5\,\text{MeV}, \qquad m_d \sim 10\,\text{MeV}, \tag{1.21}$$

$$m_s \sim 100\,\text{MeV}, \qquad m_c \sim 10^3\,\text{MeV}, \tag{1.22}$$

$$m_b \sim 5\,\text{GeV}, \qquad m_t \sim 170\,\text{GeV}. \tag{1.23}$$

So the lecture presses on a tension that is central to the Standard Model. The quarks are in one sense copies of one another; in another sense their mass scales wander across several orders of magnitude. That repetition of families without any satisfying explanation is one of the deep puzzles of the subject.

The lecture also restores some phenomenological perspective. Of the six quark flavors, only u and d appear abundantly in ordinary matter. The heavier ones decay down the chain. So low-energy physics is dominated by the first family, while the others are present as heavier echoes.

1.6.1 Question & Answer

Question. Why does a quark carry baryon number 1/3, and can baryon number be violated?

Answer. Baryon number is introduced operationally, not metaphysically. One first counts protons and neutrons. By convention,

$$B(p) = 1, \qquad B(n) = 1. \tag{1.24}$$

That normalization could have been chosen differently; the lecture explicitly remarks that one could have called it 2, or 7, or π, as long as it scaled consistently. Once the proton is recognized as a three-quark object, however, consistency forces

$$B(q) = \frac{1}{3}, \qquad B(\bar{q}) = -\frac{1}{3}. \tag{1.25}$$

So the factor 1/3 is not a new mystery. It is the inherited normalization of a bookkeeping convention.

The sharper question is whether baryon number is exact. Empirically it is conserved to extremely high precision, but the lecture refuses to promote that fact into an exact law. A decay such as

$$p \to e^+ + \gamma \tag{1.26}$$

would conserve electric charge and still violate baryon number. No such process has been observed; the proton lifetime must be at least of order 10^{33} to 10^{34} years. But that does not amount to a proof that baryon number is absolute. The lecture leaves the question open.

A further caution is also preserved. Quarks are not observed as isolated free particles. One can certainly detect evidence of quarks, but not a lone quark sitting in isolation waiting to be measured.

1.7 From Quarks to Hadrons: Nucleons, Strange Baryons, and Mesons

Once the quark charges are known, familiar hadrons can be rebuilt. The first examples are the neutron and proton:

$$n = ddu, \qquad p = uud. \tag{1.27}$$

Worked example. The lecture explicitly checks the charges:

$$Q(n) = Q(d) + Q(d) + Q(u) = -\frac{1}{3} - \frac{1}{3} + \frac{2}{3} = 0, \tag{1.28}$$

$$Q(p) = Q(u) + Q(u) + Q(d) = \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = 1. \tag{1.29}$$

Now comes a characteristic Susskind move. Forget photons for a moment. If we ignore electromagnetism and focus only on the quark content, then proton and neutron look almost like the same object with u and d interchanged. This suggests a near-symmetry. It is not exact, because u and d do not have exactly the same masses.

That near-symmetry is good enough to tell us the sign of the proton-neutron mass difference. Since $m_d > m_u$, the neutron, which contains more down quarks, should be slightly heavier. And it is. Electromagnetic self-energy by itself would push the other way and make the proton heavier, because the proton is charged. The observed ordering therefore tells us that the $d - u$ mass difference wins over the electromagnetic correction.

Remark 1.1. The lecture also points out the famous mismatch in the nucleon mass budget. The quark masses add up to only a few MeV, whereas the proton and neutron masses are around 940 MeV. The missing contribution is not mysterious "stuff" hidden in the nucleon; it is energy carried by the gluonic and binding dynamics. The lecture's summary is blunt: masses do not simply add relativistically for bound systems.

The same constituent logic produces strange baryons. Since s and d have the same electric charge, one may replace a down quark by a strange quark without changing the charge pattern:

$$uds, \qquad uss, \qquad uus. \tag{1.30}$$

These are analogs of the ordinary nucleons, but heavier, and therefore unstable. They decay down to lighter baryons plus other particles. The lecture is not yet trying to classify the decays in detail; it is simply establishing that the strange quark may be inserted into the same three-quark structures.

At this point the lecture inserts a useful rule of thumb. Why do three quarks bind into baryons? Why not two quarks? In the early days of quark theory this was a genuine puzzle. At the level of this lecture no full derivation is offered. Instead one adopts the empirical rule that the bound states one actually encounters carry charges that are integer multiples of the electron charge. Baryons do so with three quarks; mesons do so with quark-antiquark pairs. The lecture explicitly treats this as a working rule before deeper explanations are taken up.

Mesons are therefore introduced next as $q\bar{q}$ composites. Their baryon number vanishes:

$$B(q\bar{q}) = \frac{1}{3} - \frac{1}{3} = 0. \tag{1.31}$$

The first mesons to discuss are the ones built only from u and d, because those are the light quarks: they are abundant, comparatively stable, and cheap to produce.

The natural flavor basis is

$$|\bar{u}d\rangle, \qquad |u\bar{d}\rangle, \qquad |\bar{u}u\rangle, \qquad |\bar{d}d\rangle. \tag{1.32}$$

The charged states are immediate:

$$|\pi^-\rangle = |\bar{u}d\rangle, \qquad |\pi^+\rangle = |u\bar{d}\rangle. \tag{1.33}$$

1.7.1 Question & Answer

Question. Why are the neutral meson states written as superpositions rather than as separate $\bar{u}u$ and $\bar{d}d$ particles?

Answer. Because at this point the lecture is explicitly doing quantum mechanics. The states $|\bar{u}u\rangle$ and $|\bar{d}d\rangle$ are a basis, not automatically the physical particles. One may superpose them, and in fact the lecture says that neither one by itself corresponds to a distinct particle. The two orthogonal combinations are

$$|\pi^0\rangle \approx \frac{|\bar{u}u\rangle - |\bar{d}d\rangle}{\sqrt{2}}, \tag{1.34}$$

$$|\eta\rangle \approx \frac{|\bar{u}u\rangle + |\bar{d}d\rangle}{\sqrt{2}}. \tag{1.35}$$

This is also the lecture's first glimpse of isospin. Upness and downness are treated as a two-state internal degree of freedom, analogous in a loose way to spin up and spin down. In that language the neutral combinations are not arbitrary. They are the natural sum and difference states in the two-dimensional flavor space.

The pion triplet π^+, π^0, π^- goes together because the three masses are close. In the limit $m_u = m_d$, they would be even closer, ideally degenerate. The orthogonal neutral state is the η, which behaves similarly in many respects but sits higher in mass:

$$m_\pi \approx 140\,\text{MeV}, \qquad m_\eta \approx 500\,\text{MeV}. \tag{1.36}$$

The lecture treats the anomalous heaviness of the η as an important but more complicated story, not to be unpacked in this first chapter.

A further empirical remark from the lecture is worth keeping: the pions are comparatively long-lived because they do not decay by the strong interaction. Their decays proceed only through weaker channels.

Finally, once the strange quark is admitted, one obtains the strange mesons or K-mesons by replacing d or $\bar{d}$ with s or $\bar{s}$. The flavor states are

$$\bar{u}s, \qquad u\bar{s}, \qquad \bar{d}s, \qquad d\bar{s}. \tag{1.37}$$

Their charge assignments are

$$Q(\bar{u}s) = -1, \qquad Q(u\bar{s}) = +1, \tag{1.38}$$

$$Q(\bar{d}s) = 0, \qquad Q(d\bar{s}) = 0. \tag{1.39}$$

So there are two charged K-mesons and two neutral ones.

Counting all light u, d, s quarks and antiquarks gives

$$3 \times 3 = 9 \tag{1.40}$$

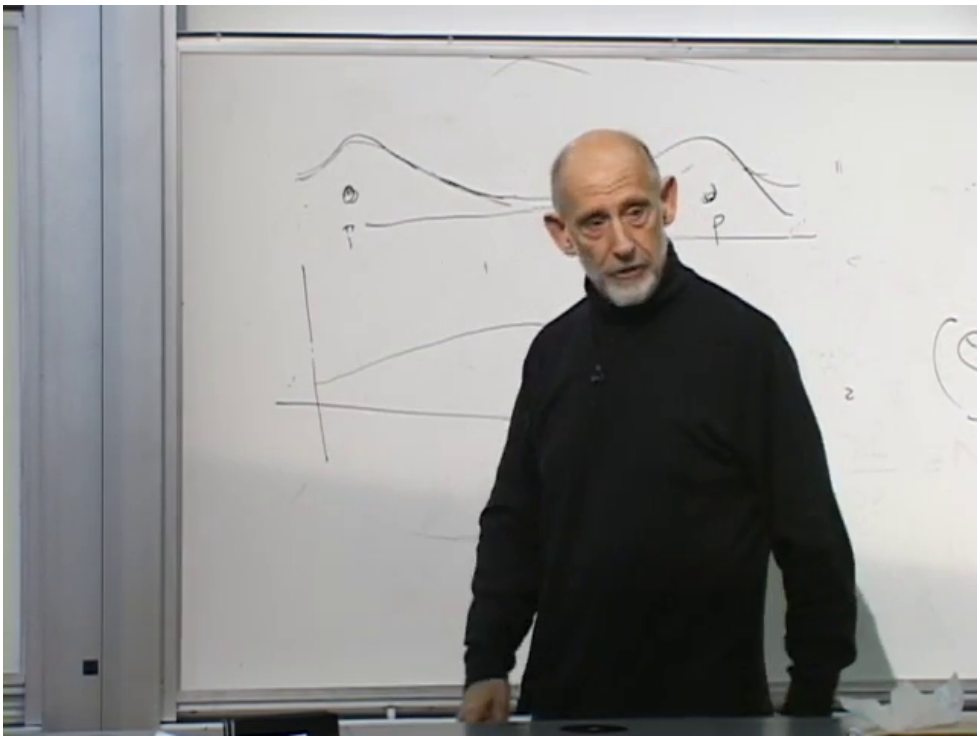
light flavor combinations in all. The lecture stops there, deliberately before complete classification. The first map of the territory has been drawn, but only in outline. That is exactly the right stopping point for Lecture 1.

1.8 Summary

The lecture begins by postponing the particle list and ends by justifying that postponement. We first see, in classical electrodynamics, that force can be read from the part of the field energy that depends on configuration. We then see the same idea retold in a different language through molecular exchange: two centers, tunneling, superposition, lowered energy, and therefore attraction.

Only after that bridge is built does the lecture shift into catalogue mode. The photon and its field A, the electron and its field ψ_e, the QED vertex, then quarks, baryon number, replicated families, nucleons, strange baryons, and neutral meson superpositions. The Standard Model is not presented as a finished elegant structure. It is presented, more honestly, as a partly organized and partly mysterious collection of particles whose patterns are visible long before their deepest explanation is in hand.

That is the right tone for the beginning of the subject. The lecture does not merely list what exists. It shows us how the concepts fit together well enough that the list can begin to mean something.



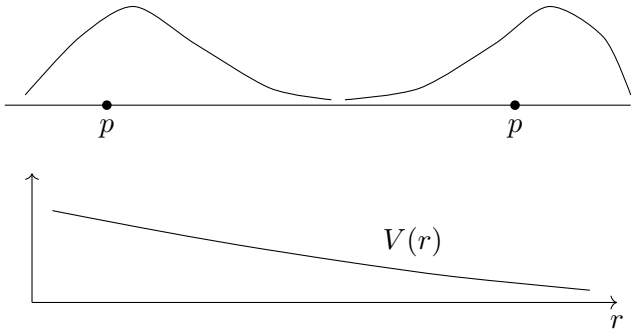


Figure 1.3: Wave-function overlap and effective proton potential. The board image supports the two-level layout; the label $V(r)$ is a cleaned choice, since the frame itself does not fix whether the intended ordinate was E or V.

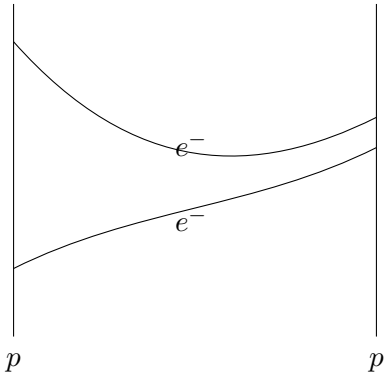


Figure 1.4: Transcript-backed schematic of electron exchange between two proton worldlines. This is an explanatory reconstruction of the lecture's picture, not a direct board transcription.

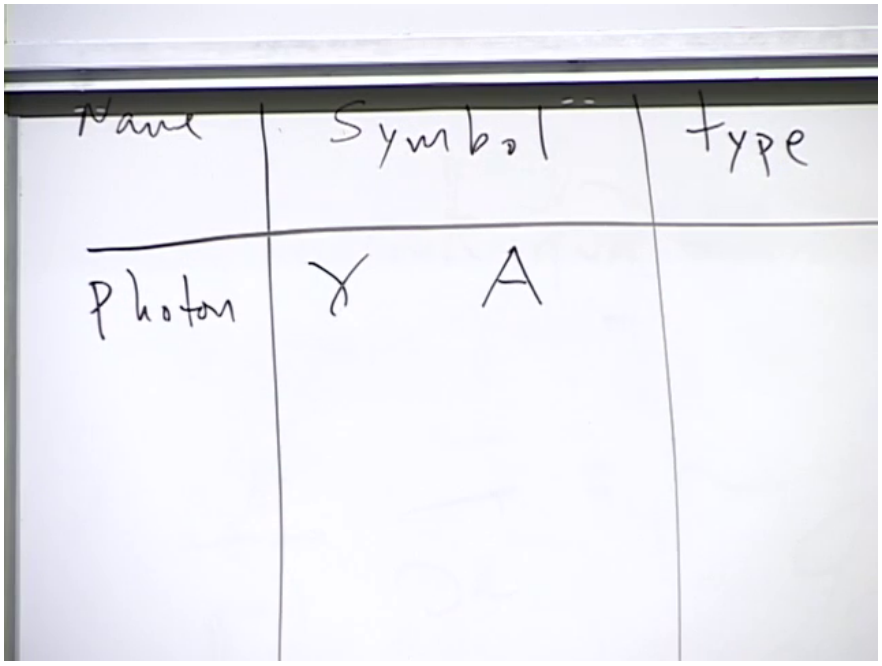


Figure 1.5: Photon symbol and vector-potential notation. The crop securely shows the photon row and the entries γ and A.

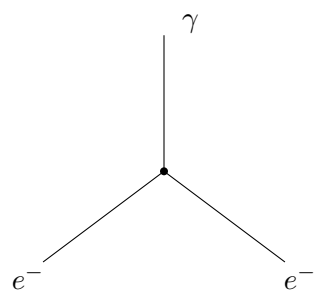


Figure 1.6: Schematic QED vertex: one electron in, one electron out, one photon attached. By crossing the legs, the lecture reads the same vertex in several ways.

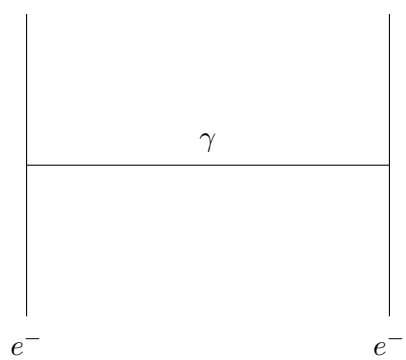


Figure 1.7: Transcript-backed schematic of photon exchange. In the lecture this is the quantum-field-theoretic retelling of the Coulomb interaction.

Chapter 2

Lecture 2: New Revolutions in Particle Physics: Standard Model

These notes follow Leonard Susskind's second lecture in the supplementary Standard Model sequence, with curation by LazyingArt LLC. The lecture begins with a warning and a promise. Quantum chromodynamics is far too large a subject for a single meeting, so we will not attempt its deep formal mathematics here. What we will do instead is extract the basic picture. The path is deliberate: QCD is first framed by analogy with QED, then the lecture backs up into the mathematics of spin, and only after that uses the same machinery to introduce isospin, the Δ puzzle, color, gluons, and the first qualitative form of QCD.

2.1 QCD by Analogy, and Why We Must Revisit Spin

Susskind opens by placing quantum chromodynamics beside quantum electrodynamics. QED is the theory of electrons and photons; in the atomic setting it is also the theory of the Coulomb force, the field that binds electrons to nuclei. In the same sense, QCD is the theory of quarks and gluons, and of the composite things that quarks and gluons make. Those composites are hadrons. Protons and neutrons are baryons, with net baryon number; mesons are quark–antiquark pairs; and in either case the gluons are part of the binding medium. The lecture wants us to see that QCD is not merely a list of particles. It is a theory of constituents and of the force that holds them together.

But before QCD proper begins, the lecture steps backward. Why? Because the mathematics of spin will come back twice: first almost directly as isospin, and then again, in a looser but still structurally related way, as color. The lecture explicitly refuses a group-theory treatment. We are told that group theory is really lurking underneath, but the working language will be angular momentum, because that language is already familiar and already rich enough for the combinatorics we need.

2.2 Angular Momentum Multiplets and Two Spin-Half Particles

The reminder begins with the commutator algebra of angular momentum:

$$[L_x, L_y] = i\hbar L_z, \tag{2.1}$$

together with the cyclic companions

$$[L_y, L_z] = i\hbar L_x, \qquad [L_z, L_x] = i\hbar L_y. \tag{2.2}$$

The lecture immediately emphasizes what this means operationally. The physics is rotationally symmetric, but our description does not have to display that symmetry at every step. We choose a preferred axis, conventionally the z-axis, and we ask for the component of angular momentum along that axis. This is not symmetry breaking in nature; it is only the expedient choice that makes the representation manageable. Since the components do not commute, we cannot measure them all at once. So we focus on one of them.

From this point of view a spin multiplet is organized by the largest value attained by the z-component. The lecture says L aloud, while the blackboard uses ℓ; to avoid confusion with the operators L_x, L_y, L_z, we will write the multiplet label as ℓ. The familiar counting rule is then

$$m_z = \ell, \ell - 1, \ldots, -\ell, \qquad \#\text{ of independent states} = 2\ell + 1. \tag{2.3}$$

That is the key fact the lecture wants us to carry forward.

Now let us put together two spin-half particles, ignoring orbital angular momentum entirely. The product basis is

$$|\uparrow\uparrow\rangle, \qquad |\uparrow\downarrow\rangle, \qquad |\downarrow\uparrow\rangle, \qquad |\downarrow\downarrow\rangle. \tag{2.4}$$

The maximum m_z is $+1$, attained when both spins point up, and the minimum is -1, attained when both point down. But there is already something to notice: there are two distinct basis states with $m_z = 0$, namely $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$. This is precisely where the lecture stops being a review and becomes a problem.

The board frame is useful not only because the mixed-spin sum has been circled, but also because the earlier spin algebra still remains visible at the top of the board. One can literally see the lecture passing from the commutator and the $2\ell+1$ counting rule to the first nontrivial example. In cleaned notation the circled state is

$$\frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}. \tag{2.5}$$

The factor $1/\sqrt{2}$ is the standard normalized completion. It is not yet visible in the earlier frame, but the transcript makes clear that this is the next move in the lecture.

2.2.1 Question & Answer

Question. If two spin-half particles give four basis states, why do they not form a single spin-one multiplet?

Answer. Because a spin-one multiplet has only three states:

$$m_z = +1, \quad 0, \quad -1.$$

The two aligned states already account for the $m_z = \pm 1$ members. So a spin-one multiplet can contain only one linear combination from the two-dimensional $m_z = 0$ sector. There is still one orthogonal combination left over. It cannot be spin two, because that would require five states altogether. It cannot be anything half-integral, because the total spin of two spin-half particles is integral. The only remaining possibility is spin zero.

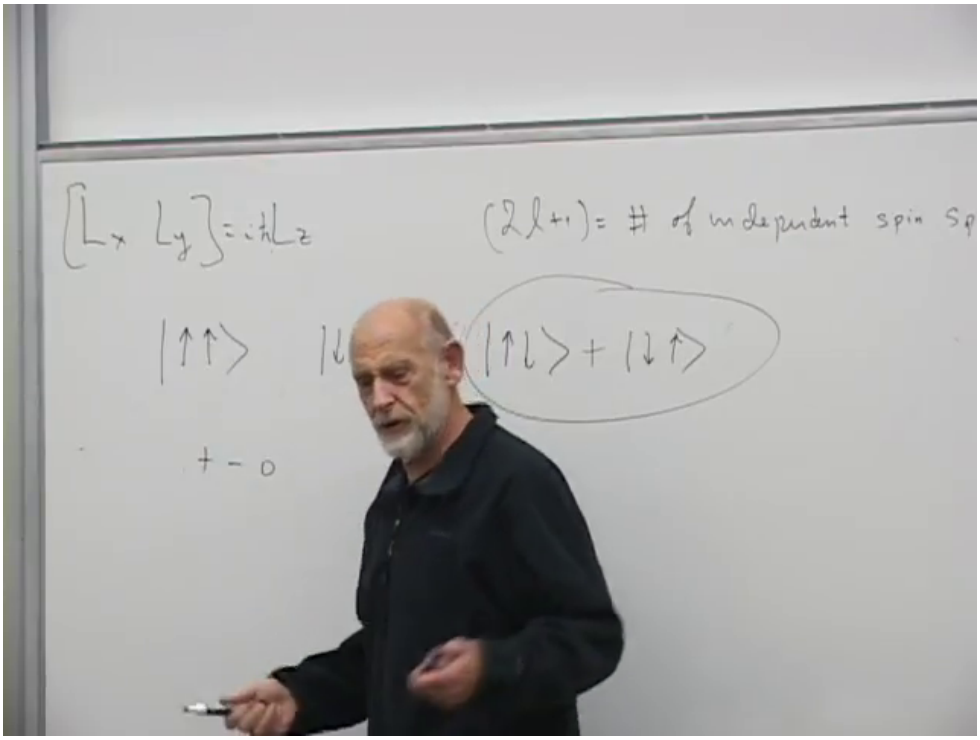


Figure 2.1: Symmetric mixed-spin state highlighted on the board. The frame preserves the moment at which the mixed-spin sum is singled out before the normalization is written explicitly.

2.3 Triplet, Singlet, and the Meaning of the Symmetric State

The lecture now resolves the counting puzzle exactly in that way. We diagonalize the $m_z = 0$ subspace into symmetric and antisymmetric combinations, and the four product states reorganize themselves as

$$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0, \tag{2.6}$$

$$\ell = 1: \qquad |\uparrow\uparrow\rangle, \qquad \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}, \qquad |\downarrow\downarrow\rangle, \tag{2.7}$$

$$\ell = 0: \qquad \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}. \tag{2.8}$$

This is a familiar identity, but in the lecture it is not merely quoted. It is forced by counting, and then interpreted physically.

The essential interpretive point is the one Susskind presses hardest: the symmetric mixed-spin state has $m_z = 0$, but it is not spin zero. It belongs to the $\ell = 1$ triplet. That is why it rotates into the $m_z = \pm 1$ states when the axis is changed. Zero along the chosen axis is not the same thing as zero total angular momentum. Only the antisymmetric combination has total angular momentum zero:

$$\frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}} \quad \longleftrightarrow \quad \ell = 0. \tag{2.9}$$

The lecture then sharpens the distinction with a physical test. Suppose we ignore every conservation law except angular momentum. Which of these states could simply disappear? The aligned states

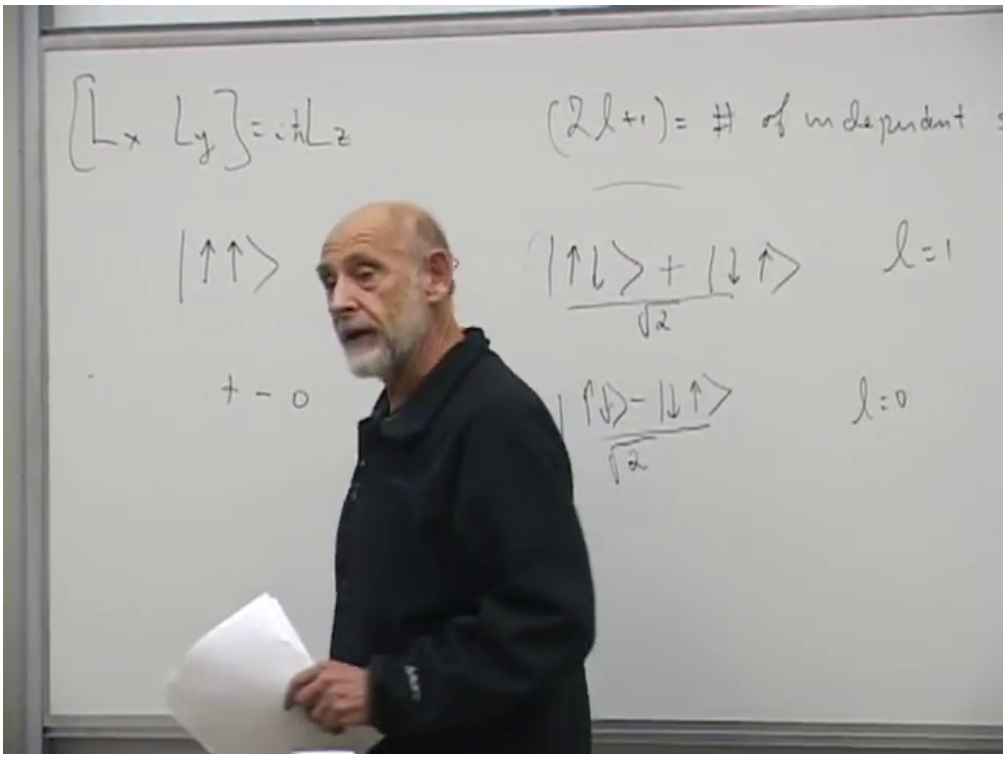


Figure 2.2: Triplet and singlet decomposition of two spin-$\frac{1}{2}$ states. The board layout itself matters: the three $\ell = 1$ states are grouped together, while the antisymmetric $\ell = 0$ state is set apart below.

clearly cannot. The symmetric $m_z = 0$ state also cannot, because although its z-component vanishes, it still belongs to the spin-one multiplet. The only combination that could disappear on angular-momentum grounds alone is the antisymmetric singlet. In that sense the singlet behaves, as Susskind says, like empty space as far as angular momentum is concerned.

This also clarifies a question from the room. If we measure the symmetric mixed-spin state along the z-axis, we obtain zero. But rotate the axis, and the same state reveals the nonzero angular momentum appropriate to the triplet. The lecture wants us to keep that distinction in mind, because exactly this sort of counting-with-interpretation will reappear when spin is replaced by isospin.

2.4 Three Spins as the Template for Isospin

At this point the lecture pauses and asks, in effect, why we are doing all this. Before answering, it runs the same counting argument once more for three spin-half particles. There are

$$2 \times 2 \times 2 = 8 \tag{2.10}$$

product states altogether. If all three spins point up, the maximum m_z is $+\frac{3}{2}$, so a spin-$\frac{3}{2}$ multiplet must be present. That multiplet contains

$$2\left(\frac{3}{2}\right) + 1 = 4 \tag{2.11}$$

states. Four states therefore remain. They cannot form another spin-$\frac{3}{2}$ multiplet, because the $\pm\frac{3}{2}$ extrema have already been used. So the only possibility is spin-$\frac{1}{2}$, and since four states remain,

spin-$\frac{1}{2}$ appears twice:

$$\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}. \tag{2.12}$$

The lecture's phrase is: let us keep that in mind.

Now the reason for the spin review arrives. What Susskind really wants to discuss is isospin, the first important internal analogue of spin in particle physics. It does not refer to the rotation of physical space. It refers to a two-state internal structure, mathematically isomorphic to the spin-half system. The secure point, especially where the transcript becomes noisy, is that isospin is an internal copy of the same mathematics, not a literal spatial spin.

The justification is physical, not merely formal. The natural scale of hadron physics is a few hundred MeV. On that scale the up and down quarks are both very light, and, more importantly, nearly degenerate. Electromagnetic effects also distort the symmetry only slightly in the hadronic setting we are considering. So as a first approximation we may neglect the mass difference between u and d and ignore their electric charges. Then the pair

$$u, \qquad d$$

behaves as a doublet under an approximate internal symmetry. In that approximation we may treat the up and down quarks as the analogue of spin up and spin down.

Historically isospin predates quarks; the lecture briefly recalls that it goes back to Heisenberg and to the older interchange of proton and neutron. But here it is introduced in the modern order: the light quark pair provides the internal two-state system, and from there the old isospin language becomes natural again.

2.5 From Spin to Isospin: Up and Down Quarks, Proton and Neutron

Once the u–d doublet is in place, the next step is almost forced. A single quark is not the simplest object we observe in the laboratory. The simplest hadrons in the discussion are three-quark states, namely the proton and the neutron. By direct analogy with the three-spin problem, three light quarks can make an isospin-$\frac{1}{2}$ sector and an isospin-$\frac{3}{2}$ sector. The lecture chooses to begin with the milder case, the isospin-$\frac{1}{2}$ doublet.

The two states are

$$p \sim uud, \qquad n \sim udd. \tag{2.13}$$

This is the hadronic counterpart of the spin-half multiplet. Two states only, related by the interchange $u \leftrightarrow d$.

Their charges follow from

$$Q(u) = +\frac{2}{3}, \qquad Q(d) = -\frac{1}{3}. \tag{2.14}$$

Thus

$$Q(p) = \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1, \qquad Q(n) = \frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0.$$

The small mass difference between proton and neutron is one of the lecture's practical signs that the symmetry is only approximate, but still good enough to organize the strong-interaction multiplets.

At this point the lecture briefly becomes more precise than it wants to remain. The quarks are labeled by positions $1, 2, 3$, and the board writes an antisymmetrized factor involving the two up quarks:

$$d_1(u_2u_3 - u_3u_2). \tag{2.15}$$

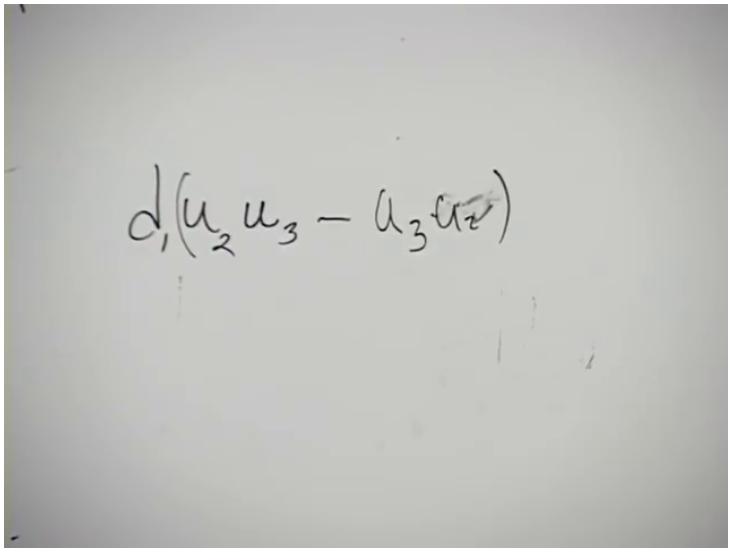


Figure 2.3: Antisymmetrized two-u-quark factor in the proton state. This is documentary evidence for the board expression only; it is not the full proton wavefunction.

The lecture immediately tells us not to lean too hard on this formal step. It is being precise, but only locally. The board shows one antisymmetrized factor, not the full proton wavefunction. What matters is the analogy with the two-spin singlet: an antisymmetric pair behaves like the zero-angular-momentum combination in the earlier discussion, and together with the third quark one arrives at an overall isospin-$\frac{1}{2}$ state. Susskind then says, in effect, never mind the fancy part; the important point is that the proton and neutron form the isospin-half doublet.

If one wished to normalize the displayed two-u-quark factor in the usual way, one would insert a factor of $1/\sqrt{2}$, but that factor is not written on the board and is not needed here. The chapter should preserve the actual board evidence and no more.

2.6 The Delta Multiplet and the Pauli Puzzle

Now the lecture shifts to the stronger case. If three light quarks can make isospin $\frac{1}{2}$, they can also make isospin $\frac{3}{2}$. The analogue of the spin-$\frac{3}{2}$ quartet is the Δ multiplet:

$$\Delta^{++} \sim uuu, \qquad \Delta^{+} \sim uud, \qquad \Delta^{0} \sim udd, \qquad \Delta^{-} \sim ddd. \tag{2.16}$$

All four states are related by the approximate isospin symmetry and therefore have nearly the same mass, up to the small effects of the u–d mass difference.

At the same time these states are not just the proton and neutron in disguise. Their ordinary spin is different. The proton and neutron have actual spin $\frac{1}{2}$; the Δ states have spin $\frac{3}{2}$. In the lecture this means that the three ordinary spins are aligned, and the three isospins are aligned as well. The

quartet therefore makes very concrete the combinatorial pattern that we first met in the abstract three-spin problem.

Their charges are

$$Q(\Delta^{++}) = +2, \qquad Q(\Delta^{+}) = +1, \qquad Q(\Delta^{0}) = 0, \qquad Q(\Delta^{-}) = -1. \tag{2.17}$$

The lecture quotes the proton and neutron mass scale as roughly 940 MeV and the Δ mass scale as roughly 1200 MeV. That is enough to make the Δ unstable.

A worked example is the decay of the doubly charged member. Start from

$$\Delta^{++} \sim uuu. \tag{2.18}$$

As the quarks pull apart, a quark–antiquark pair may appear. The lecture first toys with a $u\bar{u}$ insertion, then rejects it on energetic grounds and chooses $d\bar{d}$. That gives

$$uuu \longrightarrow uud + u\bar{d} \tag{2.19}$$
$$= p + \pi^{+}. \tag{2.20}$$

Charge conservation is immediate:

$$Q(p) = +1, \qquad Q(\pi^{+}) = Q(u) + Q(\bar{d}) = \frac{2}{3} + \frac{1}{3} = +1.$$

So the final state again has total charge +2. The mass balance is also consistent with the lecture's rough numbers:

$$1200 \text{ MeV} > 940 \text{ MeV} + 140 \text{ MeV},$$

so the decay has phase space available. This is why the Δ states are real but very short-lived resonances.

2.6.1 Question & Answer

Question. How can a Δ^{++} made of three up quarks avoid violating Fermi statistics?

Answer. This is the central tension of the lecture. In Δ^{++} the three quarks are all u, and because the baryon has spin $\frac{3}{2}$, their ordinary spins are aligned as well. So if spin and flavor were the whole story, we would seem to have three identical fermions in the same state. That is forbidden by the Pauli principle. The lecture presents this not as a minor technical issue but as a historical contradiction that had to be resolved. The resolution is that quarks must carry one more label, hidden at first from view, so that the three quarks in the Δ^{++} are not identical in every respect. That extra label is color.

The lecture even pauses to note why the Δ was so important historically. In the proton and neutron one could imagine the problem being disguised by more complicated symmetrization. In the Δ^{++} the contradiction is blunt: three up quarks, parallel spin, parallel isospin, and something is plainly wrong unless a new quantum number is introduced.

2.7 Color, Gluons, and the Minimal Picture of QCD

Color now enters as the hidden label demanded by the Δ puzzle. The lecture uses the conventional names

$$r, \qquad g, \qquad b,$$

with corresponding anti-colors

$$\bar{r}, \qquad \bar{g}, \qquad \bar{b}.$$

The names are arbitrary. They do not refer to visible color, and the three colored versions of a given quark have the same mass and the same obvious properties. But they are distinct states, and that distinction is what rescues Fermi statistics.

Here the lecture makes a historical analogy that is worth preserving. Spin itself had once played this role in atomic physics: when two electrons seemed to occupy the same orbital state, the new quantum number of spin resolved the Pauli tension. Color plays the same conceptual role a second time. The strategy is the same: attach another label so that the particles are not really identical in all degrees of freedom. In the Δ^{++}, one red, one green, and one blue quark are enough to remove the contradiction.

The lecture then pivots from the existence of color to the dynamics built from it. Before doing so, it reminds us how economical QED is. The whole theory can be summarized by one basic vertex: an electron emits or absorbs a photon. For bookkeeping purposes one may even pretend that a photon behaves in some respects like an electron–positron pair. Not because it is literally made of one, but because it shares the neutral charge bookkeeping and the spin-one structure. That fiction is not especially useful in QED, but it becomes useful in QCD.

A quark is therefore organized as a three-component color state,

$$q = \begin{pmatrix} r \\ g \\ b \end{pmatrix}, \qquad \bar{q} = \begin{pmatrix} \bar{r} & \bar{g} & \bar{b} \end{pmatrix}. \tag{2.21}$$

A gluon is then introduced, at the lecture's level, as a color–anticolor bookkeeping object:

$$q\,\bar{q} \sim \begin{pmatrix} r\bar{r} & r\bar{g} & r\bar{b} \\ g\bar{r} & g\bar{g} & g\bar{b} \\ b\bar{r} & b\bar{g} & b\bar{b} \end{pmatrix}. \tag{2.22}$$

This suggests nine gluons at first sight. The lecture then immediately adds the caveat that one diagonal superposition is not an independent state. The full group-theoretic derivation is postponed, but the minimal cleaned statement is

$$3 \otimes \bar{3} = 8 \oplus 1. \tag{2.23}$$

The lecture's phrasing is more heuristic: it looks like nine, but really one particular diagonal sum is "a nothing." That is enough for the present level.

The primitive quark–gluon interaction is the color-changing emission process

$$q^a \longrightarrow q^b + g^{a\bar{b}}. \tag{2.24}$$

For example,

$$r \longrightarrow g + (r\bar{g}). \tag{2.25}$$

This is the QCD analogue of photon emission in QED, but now the emitted quantum itself carries color bookkeeping labels. That is why the lecture keeps emphasizing the rule: follow the lines. Track the color through the diagram. If a line turns around in time, reinterpret it as an antiparticle. At this level that is essentially the whole computational grammar.

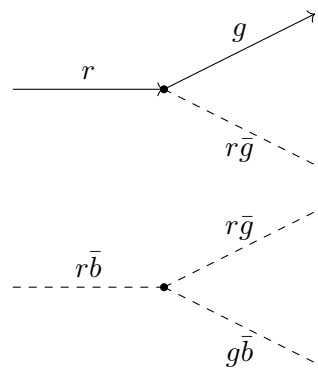


Figure 2.4: Schematic QCD vertices in the spirit of the lecture. Above: a quark changes color and emits a gluon. Below: a gluon turns into two gluons. The labels are bookkeeping labels, and the operative rule is to follow the color lines.

The upper diagram is the direct analogue of the QED vertex. The lower diagram is the new ingredient. Because photons do not carry electric charge, they do not interact with one another at the primitive level. Gluons do. The lecture constructs this by treating the gluon as a quark–antiquark-like color flow and allowing that flow itself to split. The result is a three-gluon vertex. Once that exists, gluons can be exchanged not only between quarks but also between gluons.

That is the decisive novelty of QCD. Quarks bind by exchanging gluons, but gluons themselves can also bind through gluon exchange. The lecture emphasizes three consequences. First, QCD can contain objects with no quarks at all, only gluonic degrees of freedom. Second, the dynamics of the gluon field is non-linear: one gluon wave does not simply pass through another without effect, and even different parts of a single gluon field configuration can act back on one another. Third, the bookkeeping is more intricate than in electrodynamics, which is why the lecture does not try to canonize every color-labeled sketch on the board. The stable lesson is the line-following rule, not every transient chalk label.

The lecture closes with a heuristic coda on running parameters. Imagine a small mass m attached to a soft extended blob. Push very slowly, and the whole structure moves together; the effective inertia is the mass of the full composite. Shake very rapidly, and the extended part does not have time to respond; then one probes only the small core. The observed mass depends on the scale of the probe. This is the lecture's route to the statement that the effective parameters of a quark are scale dependent:

$$m_q = m_q(\mu), \qquad Q_q = Q_q(\mu). \tag{2.26}$$

This is not a renormalization-group derivation. It is a physical analogy, meant only to explain why the "mass of a quark" and even the effective charge one attributes to it depend on how hard one probes the hadron.

The lecture deliberately refuses to close the subject here. It tells us instead what comes next: confinement, hadron structure, and the deeper reason color is never seen in isolation. For the moment the lesson is more modest. QCD works by color, gluons are the carriers of that interaction, and unlike photons they interact with one another.

2.8 Summary

The chapter has followed the lecture's intended motion from analogy to contradiction to resolution. QCD was first introduced by comparison with QED, but the real conceptual preparation was the return to spin. From the commutators and the $2\ell + 1$ rule we were led to the two-spin puzzle, then to the triplet and singlet, then to the three-spin decomposition. Only after that did the lecture reinterpret the same mathematics as isospin.

With isospin in hand, the proton and neutron emerged as the isospin-$\frac{1}{2}$ doublet, while the Δ quartet supplied the isospin-$\frac{3}{2}$ sector and, more importantly, the Pauli puzzle that forces color into the theory. Once color is admitted, the lecture can finally pass to QCD proper: color-labeled quarks, color–anticolor gluon states, the basic quark–gluon vertex, the three-gluon interaction, and the non-linear dynamics that distinguishes QCD sharply from electrodynamics.

The chapter therefore ends exactly where the lecture wants to end: with a mathematically guided intuitive picture, not with a completed theory of hadrons. The next step is confinement. Here the purpose was to see why color had to be invented, what kind of force carrier the gluon is, and how the first narrative of QCD is assembled.

Chapter 3

Rotations, Spinors, and the Road to Color

We begin with rotations because where we are going is color. Leonard Susskind chooses the most concrete possible road into the subject: start with a symmetry we can picture, let it act on objects we already know, and then watch the same mathematics reappear in less familiar settings. These notes follow that lecture closely, with chapter curation by LazyingArt LLC. The thread running through the whole discussion is simple and powerful: the same group-theoretic structure can act on spatial vectors, on spin states, and finally on quark color.

3.1 Rotations as the Warm-Up for Color

Rotations of space are symmetries, and one of the first things they act on is an ordinary vector. That is the point of departure. But it is not the destination. Susskind says early on where he is heading: to color, meaning the red, green, and blue states of a quark. Color is not a two-state system like spin up and spin down. It is a three-state system, and the interesting operations are the ones that mix those three states among themselves.

So the lecture does not study rotations for their own sake. It studies rotations because they give us a familiar example of a symmetry group together with its representations. Once that pattern is in hand, the same logic can be lifted from space to spin and from spin to color.

3.2 Axis, Angle, and the Group of Rotations

How do we parameterize a rotation? Susskind begins in the most geometric language. A rotation has an axis, which we describe by a unit vector $\hat{n}$, and it has an angle θ through which we rotate about that axis. The unit vector requires two parameters, which one may picture as longitude and latitude on the sphere, and the angle gives one more. Thus a rotation is a three-parameter object.

The board sketch is deliberately minimal. It shows the three spatial axes, a distinguished direction $\hat{n}$, and a nearby θ. The lecture speaks of a unit vector pointing to a unit sphere, but the frame itself shows only the axis sketch and the direction arrow.

The lecture also fixes a convention: we take rotations to be right-handed. A left-handed rotation

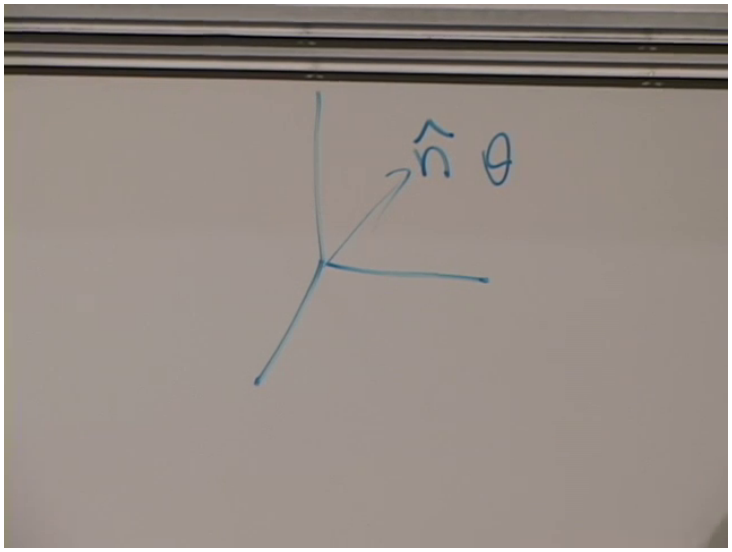


Figure 3.1: Unit-vector axis sketch for a rotation.

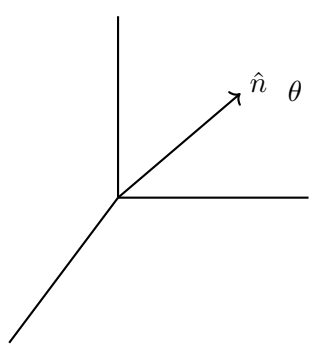


Figure 3.2: Cleaned reconstruction of the axis-angle sketch.

may be rephrased either as a right-handed rotation by a negative angle or as a right-handed rotation about the opposite axis. The point here is not to settle the global topology of the rotation group. It is only to isolate the local data $(\hat{n}, \theta)$ that carry the three parameters.

3.2.1 Question & Answer

Question. Do rotations really form a group, and what exactly makes them non-abelian?

Answer. Yes. If we rotate an object about one axis and then about another, the result is again some rotation. That gives closure. There is an identity element, namely no rotation at all. Every rotation has an inverse, obtained by rotating through the opposite angle about the same axis. Associativity also holds.

What fails is commutativity. A rotation about the x-axis followed by a rotation about the y-axis is not, in general, the same as the same two rotations performed in the opposite order. These are sequential operations, one after another. The lecture even remarks that for very small rotations the failure of commutativity becomes less dramatic, but the group remains unmistakably non-abelian. This is the first non-abelian group encountered in the course.

3.3 Vectors and the 3×3 Representation

Once the group structure is on the table, Susskind asks for a representation. The simplest one is the representation on the components of an ordinary spatial vector. Let V_i be the components of a vector V, with $i = 1, 2, 3$ or, if we prefer, $i = x, y, z$. For every rotation there is a corresponding 3×3 matrix $R_{ij}(\theta, \hat{n})$ such that

$$V_i' = R_{ij}(\theta, \hat{n})\, V_j. \tag{3.1}$$

This is the bridge from geometry to algebra. The board writes the equation in the informal order

$$R_{ij}(\theta, \hat{n})\, V_j = V_i',$$

and we standardize it as

$$V_i' = R_{ij}(\theta, \hat{n})\, V_j.$$

For the moment the vector components and the matrix entries are taken to be real.

Now let us do what the lecture does on the board: let us see what invariant length says. A true rotation does not change the length of a vector, so we require

$$V_i V_i = V_i' V_i'. \tag{3.2}$$

Substituting the transformation law gives

$$V_i' V_i' = (R_{ij} V_j)(R_{ik} V_k) \tag{3.3}$$

$$= R_{ij} R_{ik} V_j V_k. \tag{3.4}$$

This must agree with $V_i V_i$ for every vector V. Therefore the coefficient multiplying $V_j V_k$ must be the Kronecker delta:

$$R_{ij} R_{ik} = \delta_{jk}. \tag{3.5}$$

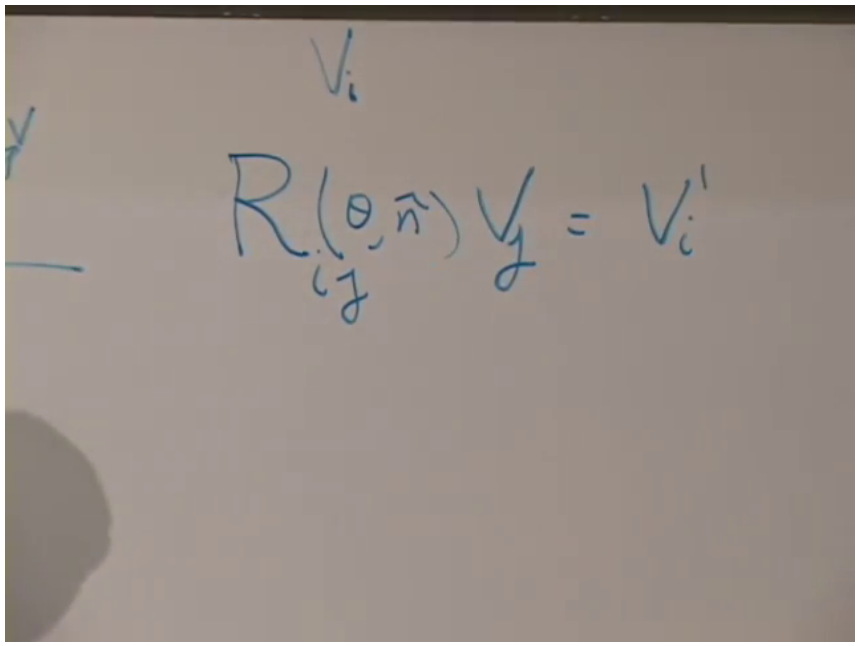


Figure 3.3: Rotation matrix acting on vector components.

In matrix language this becomes

$$R^T R = \mathbf{1}. \tag{3.6}$$

That is the defining property of a rotation matrix. It follows immediately that

$$R^{-1} = R^T. \tag{3.7}$$

This is the lecture's first real derivation, and it sets the pattern for everything that follows. We first ask what the transformation is supposed to preserve; then we translate that preservation law into a condition on the matrix representing the transformation.

It is also worth noting the broader point Susskind makes here. This 3×3 representation is not the only representation of the rotation group. It is simply the one that is easiest to visualize.

3.4 Why Spin-1 Particles Are Vector Particles

With the vector representation in hand, the lecture makes an important conceptual jump. A spin-one particle has three spin states. Therefore the state of the spin alone may be described by a three-component column vector, whose entries are the amplitudes for those three states. If we rotate the spin-one system, those three amplitudes must mix among themselves under a three-dimensional representation of the rotation group.

And in fact it is just this one. The same three-dimensional mathematics that rotates the components of a spatial vector rotates the three spin states of a spin-one particle. That is the content of the phrase that spin-one particles are vector particles. It does not mean that the state is literally a geometric arrow. It means that the three states transform the same way the three components of a vector do.

At this point the lecture pauses and takes a breath. We have already met two different kinds of groups. Some are discrete: one can either do the operation or not do it, as with a reflection. Others are continuous: the group elements depend on continuously varying parameters, as with rotations or multiplication by a phase $e^{i\phi}$. The continuous groups are the ones that will dominate the rest of the course, and they are called Lie groups.

This pause matters, because it resets the discussion just before the harder case. One can also insert here the trivial case of spin zero. A spin-zero particle is unchanged by a rotation. Its representation of the rotation group is just the identity operator and nothing more.

3.5 Spin-$\frac{1}{2}$ States and Unitary Matrices

Now we turn to the genuinely new case: spin $\frac{1}{2}$. A spin-$\frac{1}{2}$ system has two basis states,

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{3.8}$$

which we identify with spin up and spin down relative to some chosen axis, say the z-axis. A general state is

$$\psi = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}, \qquad |\alpha_1|^2 + |\alpha_2|^2 = 1. \tag{3.9}$$

The two entries are complex. So at first sight there are four real parameters. But one overall phase is physically irrelevant, and the normalization condition removes one more. That leaves two physically meaningful parameters, just enough to specify a direction. The lecture does not stop to develop that mapping explicitly; it simply uses the fact that this two-component column is the object that rotations act on.

So what does it mean to rotate such a state? The answer is the same in form as before: we act with a matrix. There is a 2×2 matrix U such that

$$\alpha'_i = U_{ij}\alpha_j. \tag{3.10}$$

The logic again begins with preservation, but now what must be preserved is not ordinary Euclidean length. It is the total probability. A state must go into another state.

Let us therefore work out the analogue of the vector derivation. We require

$$\alpha_i^* \alpha_i = \alpha_i'^* \alpha'_i. \tag{3.11}$$

Using $\alpha'_i = U_{ij}\alpha_j$, we find

$$\begin{aligned} \alpha_i'^* \alpha'_i &= \left(U_{ij}\alpha_j\right)^* \left(U_{ik}\alpha_k\right) && (3.12) \\ &= \alpha_j^* U_{ij}^* U_{ik} \alpha_k && (3.13) \\ &= \alpha_j^* \left[(U^T)^* U\right]_{jk} \alpha_k. && (3.14) \end{aligned}$$

If this is to agree with $\alpha_i^*\alpha_i$ for every state, then the matrix multiplying $\alpha_j^*\alpha_k$ must be the identity:

$$\left(U^T\right)^* U = \mathbf{1}. \tag{3.15}$$

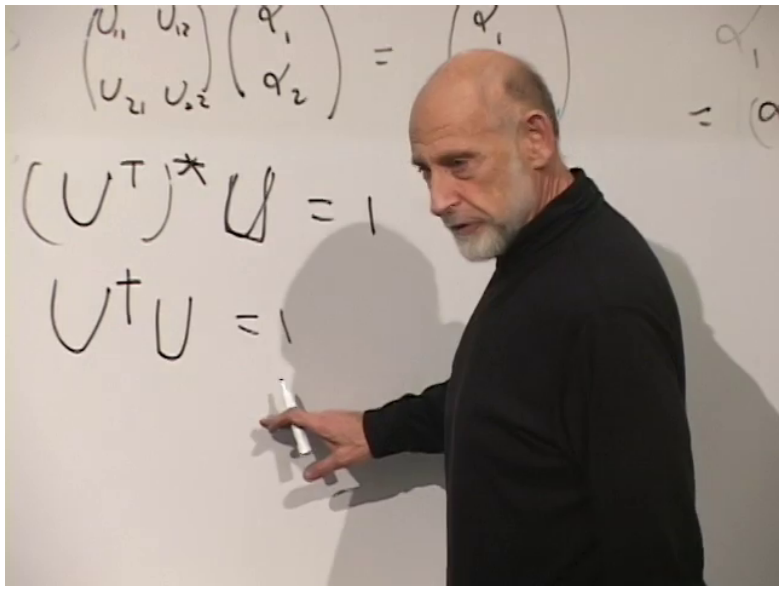


Figure 3.4: Unitary condition written in Hermitian-conjugate form.

This is exactly what appears on the board. The secure blackboard content is the pair of equations

$$\left(U^T\right)^* U = \mathbf{1}, \qquad U^\dagger U = \mathbf{1}. \tag{3.16}$$

The second line is simply the compact rewrite of the first, once we name the transpose-plus-complex-conjugate operation:

$$U^\dagger = (U^T)^*. \tag{3.17}$$

Thus the matrices that rotate a spin-$\frac{1}{2}$ state are unitary matrices.

The pattern is the same as before, but the setting is different. For vectors we preserved $V_i V_i$. For spinors we preserve $\alpha_i^* \alpha_i$. That single difference is exactly what replaces transpose by Hermitian conjugation.

3.6 From $U(2)$ to $SU(2)$, and Where the Sigmas Enter

The lecture now performs one of its characteristic parameter counts. A 2×2 complex matrix has four complex entries, hence eight real parameters. The condition

$$U^\dagger U = \mathbf{1}$$

is a matrix equation, so it gives four real constraints. Therefore a general unitary 2×2 matrix has four real parameters.

But a rotation has only three parameters. So there is one parameter too many. That is the problem the lecture wants us to feel before it introduces the next restriction.

What can that extra parameter be? Unitarity tells us only that

$$\det U \, \det U^\dagger = 1, \tag{3.18}$$

and since $\det U^\dagger = (\det U)^*$, it follows that $|\det U| = 1$. In other words, before any further restriction,

$$\det U = e^{i\phi} \tag{3.19}$$

for some phase ϕ. The lecture then fixes that phase by imposing

$$\det U = 1. \tag{3.20}$$

This removes exactly one real parameter.

Why is this consistent with the group structure? Because determinants multiply:

$$\det(AB) = \det A \det B. \tag{3.21}$$

So if two matrices both have determinant 1, their product also has determinant 1. The identity matrix has determinant 1 as well. Thus the unitary matrices with determinant 1 form a closed subgroup.

This is the special unitary group. In the notation used throughout physics,

$$U(n) : \text{unitary } n \times n \text{ matrices}, \qquad SU(n) : \text{unitary } n \times n \text{ matrices with } \det U = 1. \tag{3.22}$$

A 1×1 unitary matrix is just a phase $e^{i\phi}$, the group $U(1)$. The group acting on the two components of a spinor is $SU(2)$.

At lecture level Susskind then says something deliberately forceful: this is the rotation group. Immediately after saying it, however, he adds that there is a subtlety he is not discussing. The safe way to record the point is this: $SU(2)$ has the correct three local parameters and the correct multiplication structure for describing spatial rotations on spin-$\frac{1}{2}$ states, even though it does not act on ordinary three-vectors.

3.6.1 Question & Answer

Question. Where do the Pauli matrices come in?

Answer. They come in when we stop looking at finite transformations as whole group elements and instead examine transformations very close to the identity. Small rotations should be represented by matrices very close to $\mathbf{1}$. So let us write

$$U = \mathbf{1} + \epsilon M + O(\epsilon^2), \tag{3.23}$$

with ϵ small and real.

Now impose unitarity to first order. Since

$$U^\dagger = \mathbf{1} + \epsilon M^\dagger + O(\epsilon^2),$$

we have

$$U^\dagger U = \mathbf{1} + \epsilon(M^\dagger + M) + O(\epsilon^2). \tag{3.24}$$

For this to equal $\mathbf{1}$, we must have

$$M^\dagger = -M. \tag{3.25}$$

So in this convention the infinitesimal generator is anti-Hermitian.

Susskind then makes the standard improvement and pulls out a factor of i. Write instead

$$U = \mathbf{1} + i\epsilon M + O(\epsilon^2). \tag{3.26}$$

Then

$$U^\dagger = \mathbf{1} - i\epsilon M^\dagger + O(\epsilon^2),$$

and therefore

$$U^\dagger U = \mathbf{1} + i\epsilon(M - M^\dagger) + O(\epsilon^2). \tag{3.27}$$

Thus first-order unitarity now requires

$$M^\dagger = M. \tag{3.28}$$

So after pulling out the factor of i, the generator becomes Hermitian.

We still have to impose determinant one. Near the identity, the lecture analyzes the determinant by a direct 2×2 expansion. The clean first-order statement is

$$\det(\mathbf{1} + \epsilon M) = 1 + \epsilon \operatorname{tr} M + O(\epsilon^2). \tag{3.29}$$

Accordingly,

$$\det(\mathbf{1} + i\epsilon M) = 1 + i\epsilon \operatorname{tr} M + O(\epsilon^2), \tag{3.30}$$

and the condition $\det U = 1$ implies

$$\operatorname{tr} M = 0. \tag{3.31}$$

So the generators of small $SU(2)$ rotations are precisely the traceless Hermitian 2×2 matrices. That is exactly where the sigmas enter. The Pauli matrices span this space, so every such generator can be written as

$$M = a_x\sigma_x + a_y\sigma_y + a_z\sigma_z. \tag{3.32}$$

If the earlier spin matrices are denoted S_i, then they are proportional to these sigmas; at lecture level the only point needed here is that the sigmas furnish the basis of the infinitesimal generators. A small rotation about the x-axis uses σ_x, about the y-axis uses σ_y, and a rotation about a general axis uses the corresponding linear combination.

This is the algebraic heart of the matter. The Pauli matrices are not decorative extra objects added from elsewhere. They appear because $SU(2)$ near the identity demands traceless Hermitian 2×2 generators, and that is exactly what the sigmas are.

3.7 Color as a Three-State System and the Arrival of $SU(3)$

Only after all of this preparation does the lecture return to its announced target. A quark has three color states: red, green, and blue. So the color state of a single quark is naturally described by a three-component column,

$$q = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \qquad \text{or simply} \qquad q = \begin{pmatrix} r \\ g \\ b \end{pmatrix}. \tag{3.33}$$

The entries may be read as amplitudes, so that the squared magnitudes give the probabilities for the quark to be red, green, or blue. Susskind also remarks that one may think similarly of three field operators, one for each color, assembled into a three-component object.

One can certainly imagine discrete symmetries that interchange two colors. But the lecture wants the larger continuous family that contains such mixing in a more powerful way. The natural generalization of the spinor story is to let special unitary 3×3 matrices act on the color triplet:

$$q' = Uq, \qquad U^\dagger U = \mathbf{1}, \qquad \det U = 1, \qquad U \in SU(3). \tag{3.34}$$

This is the same mathematics as before, except that the matrices are now 3×3 rather than 2×2. It is not a rotation in ordinary space. It is a rotation in an internal three-state space.

The lecture then counts parameters just as it did for $SU(2)$. A 3×3 complex matrix has nine complex entries, hence eighteen real parameters. The condition

$$U^\dagger U = \mathbf{1}$$

gives nine real constraints at lecture level, one for each matrix entry, and the condition $\det U = 1$ removes one more. So the group has

$$18 - 9 - 1 = 8 \tag{3.35}$$

independent parameters.

This is the first moment at which the number eight appears in a serious way. Susskind immediately points out the suggestive connection: if one is wondering what this eight has to do with the eight gluons, one is wondering about the right thing. But the lecture stops short of giving the full gauge-theory construction. Even the joke that there are not nine gluons because "we threw away the determinant" is meant heuristically, as a pointer toward the eight-generator structure of $SU(3)$, not as a finished derivation.

What matters for the chapter at this stage is the symmetry statement itself. The claim is that physics is invariant under these $SU(3)$ transformations of color. That is the symmetry group of quantum chromodynamics. Just as the earlier parts of the course used symmetry to constrain what could appear in a theory, the next step will be to demand that the QCD Lagrangian be invariant under this $SU(3)$ action.

3.8 Summary

The lecture unfolds in a remarkably controlled sequence. We begin with ordinary rotations, parameterized by an axis $\hat{n}$ and an angle θ, and we verify that they form a three-parameter non-abelian group. We then represent them on vector components and derive the defining condition

$$R^T R = \mathbf{1}$$

from invariant length. That same three-dimensional representation reappears as the action of rotations on the three states of a spin-one particle.

After a brief pause to distinguish discrete from continuous symmetries, the lecture turns to spin $\frac{1}{2}$. There the state is a two-component complex spinor, and the preserved quantity is total probability rather than Euclidean length. The corresponding matrix condition is

$$U^\dagger U = \mathbf{1}.$$

Parameter counting then shows that one further restriction is needed, and $\det U = 1$ leads to $SU(2)$. Near the identity, the generators become traceless Hermitian 2×2 matrices, hence linear combinations of the Pauli matrices.

Only then do we arrive at color. A quark color state is a three-component object, and the same special-unitary logic lifts from $SU(2)$ to $SU(3)$. The resulting eight-parameter symmetry is the first mathematical sign of the gluonic structure still to come. The lecture ends exactly there: not with the full construction of QCD, but with the symmetry structure now plainly visible.

Chapter 4

New Revolutions in Particle Physics: The Standard Model

The lecture opens with a comic commercial break and a return to Alice and Bob, but the mathematical re-entry is an audience question: what exactly is the relation between an abstract group and the concrete matrices by which it acts on physical states? That is the real starting point of the chapter. From there the lecture proceeds in a very deliberate order: first rotations and $SU(2)$, then the parallel story for color $SU(3)$, then the way representations combine, and finally the dynamics that explains why only color-singlet states show up as free particles. These notes follow Leonard Susskind's lecture closely and were curated by LazyingArt LLC.

4.1 From Abstract Groups to Physical Representations

Susskind begins by insisting that we are *not finished* with group theory. The question from the room is exactly the right one: on the one hand a group looks like an abstract multiplication table, and on the other hand the lecture keeps talking about matrices acting on physical systems. Are these two different things, or are they really the same thing seen from two sides?

His answer is to take the cleanest possible example: rotations of space. A rotation is, first of all, an abstract operation. One can describe it without saying what it acts on. Physics then asks a second question: how does that abstract rotation act on the quantities that matter?

In classical mechanics the answer is familiar enough. A rotation acts on coordinates, or on the direction of a vector, or on the orientation of a magnet. In quantum mechanics the point becomes sharper, because the states of a system are represented by vectors in a linear vector space. The symmetry is abstract; the representation is the concrete matrix action on that vector space.

4.1.1 Question & Answer

Question. What is the difference between the abstract symmetry group and the matrices that act on states?

Answer. We should distinguish them, but we should not pull them apart. The group is the abstract symmetry itself, together with its multiplication law. A representation is a set of matrices that reproduces that same multiplication law on some chosen state space. The same rotation in space may therefore be represented by different matrices on different systems, but those matrices must still encode the same underlying group operation. In that sense the two notions are different, but in physics they are completely related.

4.2 Why Representation Dimension Depends on the System

Once the question is posed that way, the dimension of a representation stops being mysterious. It is fixed by the number of mutually orthogonal states of the system we are talking about.

If we keep only the spin of an electron, there are two such states, spin up and spin down, so rotations act by 2×2 matrices. If we had two such spins and ignored everything else, we would have four states. If we talk about a spin-one particle, there are three states, and rotations act by 3×3 matrices. If we go down to spin zero, rotations act trivially: the state is simply left alone.

In the language of the lecture,

$$\text{spin } 0: \ 1 \times 1, \qquad \text{spin } \frac{1}{2}: \ 2 \times 2, \qquad \text{spin } 1: \ 3 \times 3, \qquad \text{spin } \frac{3}{2}: \ 4 \times 4. \tag{4.1}$$

The one-by-one case is almost useless mathematically, but it is conceptually important: a spin-zero state is a trivial representation of rotations.

The point is not limited to spin. If we describe the full orbital motion of an electron, then there are infinitely many mutually orthogonal states. The amplitudes may be functions of position or momentum, and the corresponding representation of the rotation group is then infinite-dimensional. The abstract symmetry group has not changed; what has changed is the system on which it acts.

This is why the lecture spends time on what could otherwise look like old material. The same abstract symmetry operation can have many representations of different dimensions. That fact is the bridge from the familiar $SU(2)$ story of spin to the less familiar $SU(3)$ story of color.

4.3 Special Unitary Matrices and Infinitesimal Generators

We now come back, as Susskind says, to the spin of the electron and throw everything else out of our heads. For a spin-$\frac{1}{2}$ particle, rotations are represented by 2×2 unitary matrices U:

$$U^\dagger U = 1. \tag{4.2}$$

A general 2×2 complex matrix has four complex entries, hence eight real parameters. The unitary condition gives four real constraints. That gets us down to four parameters. But a rotation in ordinary space has only three independent parameters. The missing condition is

$$\det U = 1. \tag{4.3}$$

So for $SU(2)$ the counting is

$$8 - 4 - 1 = 3. \tag{4.4}$$

That is exactly the right number for spatial rotations.

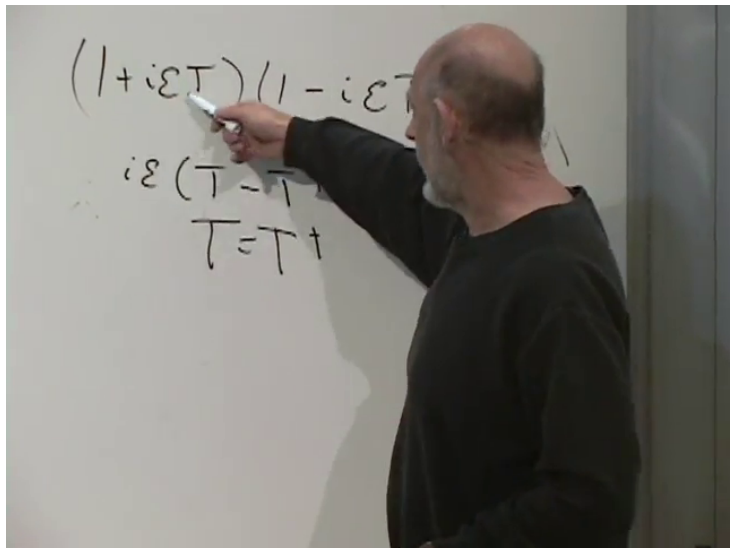


Figure 4.1: First-order unitarity implies a Hermitian generator.

The lecture pauses here to check that the determinant-one condition is consistent with multiplication. If A and B are matrices, then

$$\det(AB) = \det A \det B. \tag{4.5}$$

Also,

$$\det(U^\dagger) = (\det U)^*. \tag{4.6}$$

Taking determinants of $U^\dagger U = 1$, we find

$$(\det U)^* \det U = 1, \tag{4.7}$$

so $\det U$ must be a pure phase:

$$\det U = e^{i\theta}. \tag{4.8}$$

If we multiply U by the scalar phase $e^{-i\theta/2}$, then

$$\det\!\left(e^{-i\theta/2} U\right) = 1. \tag{4.9}$$

This isolates the special-unitary part relevant for rotations.

Here the lecture inserts an important caveat. The correspondence between spatial rotations and $SU(2)$ is not exactly one-to-one. The matrices U and $-U$ both represent the same rotation in space. So $SU(2)$ is the familiar double cover of the ordinary rotation group. The lecture does not need that fine point later, but it does not want to suppress it entirely either.

The next move is the central move of the lecture: we pass from finite group elements to infinitesimal ones. This matters because the generators are the conserved quantities. A group element very close to the identity may be written as

$$U = 1 + i\epsilon T, \tag{4.10}$$

with ϵ small. Nothing in this step is special to $SU(2)$; the same reasoning works for $SU(N)$. But here we keep the spin-$\frac{1}{2}$ picture in mind.

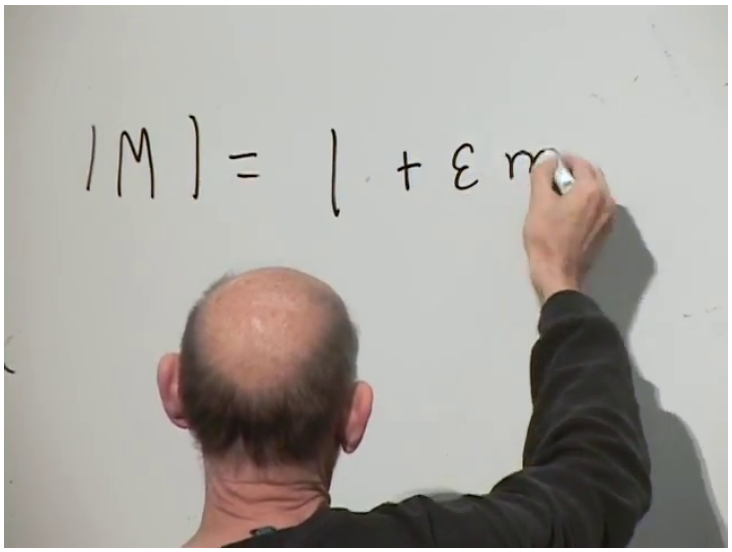


Figure 4.2: Near-identity determinant expansion on the board.

Using unitarity and keeping only terms to first order in ϵ, the board derivation is

$$(1 + i\epsilon T)(1 - i\epsilon T^\dagger) = 1, \tag{4.11}$$

$$i\epsilon\,(T - T^\dagger) = 0, \tag{4.12}$$

$$T = T^\dagger. \tag{4.13}$$

So the generator is Hermitian. That is exactly what quantum mechanics wants: Hermitian operators are observables.

At this stage the lecture asks why there are three of these generators. The reason is that the rotation group has three independent directions: infinitesimal rotations about the x-, y-, and z-axes. One can also say that the generators behave like the components of a vector. For rotations, these generators are the angular momentum operators.

Then Susskind backs up. We have used unitarity, but we have not yet used the determinant-one condition. That is where the second figure enters.

The screenshot catches the board while the determinant expansion is still being written. The fully cleaned first-order statement is

$$\det(1 + \epsilon m) = 1 + \epsilon\ \mathrm{tr}\, m + O(\epsilon^2). \tag{4.14}$$

Applying this to $U = 1 + i\epsilon T$, we get

$$\det U = 1 + i\epsilon\ \mathrm{tr}\, T + O(\epsilon^2). \tag{4.15}$$

But for $SU(2)$ we require $\det U = 1$, so the linear term must vanish:

$$\mathrm{tr}\, T = 0. \tag{4.16}$$

We have therefore arrived at the precise lecture-level characterization of the generators of $SU(2)$: they are the traceless Hermitian 2×2 matrices. There are three independent ones. In the standard

language these are the Pauli matrices, and physically they are the angular momentum operators. The lecture briefly reminds us that the whole structure of the group is encoded in the generators and, more deeply, in their commutation relations. But for present purposes the key point is simpler: the conserved quantities associated with the symmetry are the generators.

4.4 From $SU(2)$ to $SU(3)$

At this point the lecture says, half-jokingly, that we will torture Alice a little more and go on to $SU(3)$. The mathematics is intentionally repetitive: same story, bigger matrices, different physics.

A general 3×3 complex matrix has nine complex entries, hence eighteen real parameters. Unitarity gives nine real constraints, and the determinant-one condition gives one more. So

$$18 - 9 - 1 = 8. \tag{4.17}$$

Equivalently, $SU(3)$ has eight linearly independent traceless Hermitian 3×3 generators. These are the Gell-Mann matrices.

The physical interpretation now changes. A quark carries three color amplitudes, and we write its state schematically as

$$q = \begin{pmatrix} q_r \\ q_b \\ q_g \end{pmatrix}. \tag{4.18}$$

The lecture momentarily stumbles in the color names, but the stable convention is red, blue, and green. A color transformation acts by a 3×3 matrix:

$$q' = Uq. \tag{4.19}$$

Thus color rotations mix red, blue, and green exactly as spin rotations mix spin-up and spin-down amplitudes.

A pure red quark is, in that sense, the color analogue of a pure spin-up state. After an $SU(3)$ transformation it becomes a quantum superposition of red, blue, and green.

Now we ask the same question for anti-quarks. Their field operators are the complex conjugates of the quark operators, so they transform in the complex-conjugate representation:

$$\bar{q}' = U^* \bar{q}. \tag{4.20}$$

This is the $\bar{3}$, the anti-triplet. The lecture uses the $U(1)$ example to make the point intuitive: if an electron wavefunction transforms by $e^{i\theta}$, then the positron transforms by $e^{-i\theta}$. They are complex-conjugate representations. The same idea is now being used for color.

For $SU(2)$ the complex-conjugate representation is equivalent to the original one; that is a special feature of $SU(2)$. For $SU(3)$, however, the 3 and the $\bar{3}$ are distinct. The lecture states, at its own level of precision, that the generators in the $\bar{3}$ representation are the negatives of those in the 3 representation. That is the sense in which an anti-quark carries the opposite color.

4.5 How Representations Combine

Before doing color products, Susskind steps back again and asks how representations combine at all. This is not a decorative review. Without it, formulas like $3 \otimes 3 = \bar{3} \oplus 6$ would just look like mysterious multiplication tables.

Take two spin-$\frac{1}{2}$ systems and ignore all orbital motion. If we put the systems together, we get four basis states:

$$\uparrow\uparrow, \qquad \uparrow\downarrow, \qquad \downarrow\uparrow, \qquad \downarrow\downarrow\,.$$

In operator language we can think of them as built from an object with two indices, $\psi_i \phi_j$, where i and j each take the values $\uparrow$ or $\downarrow$. The $SU(2)$ action on ψ and on ϕ induces an action on this four-dimensional composite space.

The key discovery is that one particular linear combination does not mix with the others:

$$\epsilon_{ij}\psi_i\phi_j = \psi_\uparrow\phi_\downarrow - \psi_\downarrow\phi_\uparrow. \tag{4.21}$$

This is the singlet. The remaining three states mix among themselves:

$$\psi_\uparrow\phi_\uparrow, \qquad \psi_\downarrow\phi_\downarrow, \qquad \psi_\uparrow\phi_\downarrow + \psi_\downarrow\phi_\uparrow. \tag{4.22}$$

They form the triplet. Thus

$$2 \otimes 2 = 1 \oplus 3. \tag{4.23}$$

The lecture does not insist on the $1/\sqrt{2}$ normalizations, so we follow that convention here.

This is just the familiar statement that two spin-$\frac{1}{2}$ objects may combine into spin zero or spin one. The lecture even reminds us of the old atomic-language example: ortho- and para-hydrogen.

4.5.1 Question & Answer

Question. What do equations like $3 \otimes 3 = \bar{3} \oplus 6$ actually mean?

Answer. They mean that the composite state space is reducible. The left-hand side is the full tensor-product space, with all the states obtained by combining the two systems. The right-hand side says that in a suitable basis the group action splits into invariant blocks. In the $2 \otimes 2$ example, one can choose the basis consisting of the singlet and the three triplet states; then the 4×4 matrices representing the group become block diagonal, with a 1×1 block and a 3×3 block. Exactly the same logic lies behind the color formulas. The plus sign on the board is not arithmetic addition; in clean notes we write it as $\oplus$, decomposition into irreducible pieces.

With that warm-up in place, the color products begin to make sense.

For two quarks, the lecture states

$$3 \otimes 3 = \bar{3} \oplus 6. \tag{4.24}$$

The interpretation is simple and important. Two quarks give $3 \times 3 = 9$ color states. If we organize those states by symmetry under interchange of the two color indices, then the antisymmetric part has dimension 3 and furnishes the $\bar{3}$, while the symmetric part has dimension 6 and furnishes the 6.

For a quark and an anti-quark we get

$$3 \otimes \bar{3} = 1 \oplus 8. \tag{4.25}$$

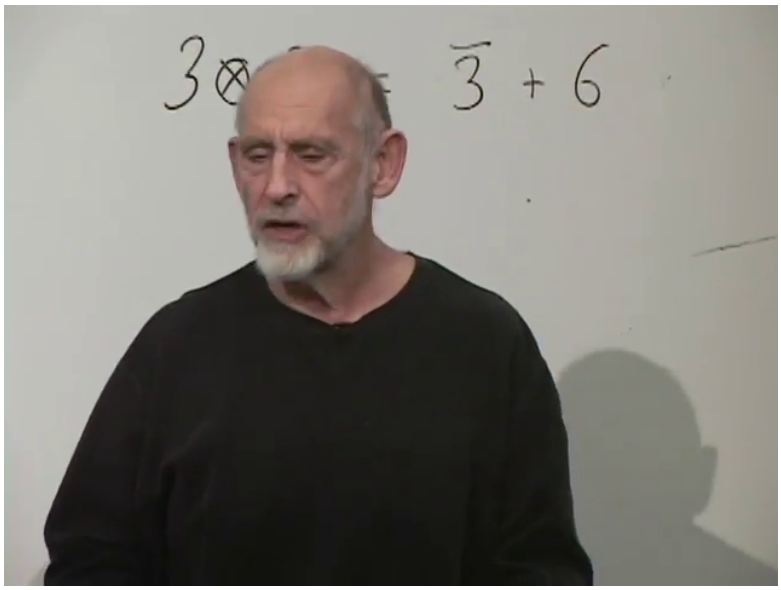


Figure 4.3: Two-quark triplet product decomposed into $\bar{3}$ and 6.

One linear combination is invariant under color:

$$q_r\bar{q}_r + q_b\bar{q}_b + q_g\bar{q}_g, \tag{4.26}$$

up to normalization. That is the singlet. The remaining eight states mix into one another and form the octet, the representation that transforms the same way as the eight generators. The lecture mentions the name *adjoint representation*, but the content is more important than the label: it is the representation carried by the gluons.

For three quarks the lecture proceeds one step at a time. First,

$$3 \otimes 3 = \bar{3} \oplus 6. \tag{4.27}$$

Then combine with the third quark. The $\bar{3} \otimes 3$ branch gives

$$\bar{3} \otimes 3 = 1 \oplus 8, \tag{4.28}$$

so

$$3 \otimes 3 \otimes 3 \supset 1. \tag{4.29}$$

The singlet is written with the antisymmetric epsilon tensor:

$$\epsilon_{ijk} q_i q_j q_k. \tag{4.30}$$

Because ϵ_{ijk} vanishes unless i, j, and k are all different, this is precisely the antisymmetrized red–green–blue combination.

The lecture briefly remarks, cautiously, that the other branch produces higher irreducible representations, for example the $6 \otimes 3$ branch leading to higher multiplets. But the singlet is the branch that matters most physically, and it is the one to which the lecture immediately returns.

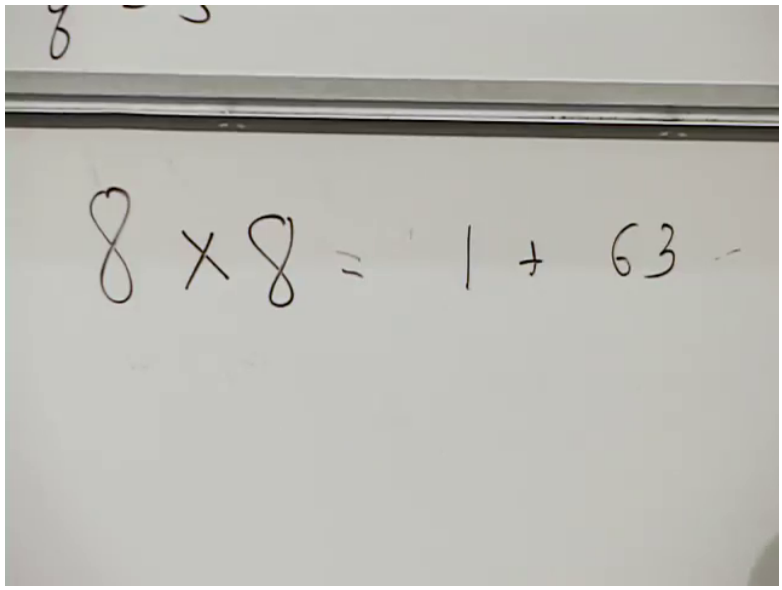


Figure 4.4: Two-gluon state count written as $8 \times 8 = 1 + 63$.

4.6 Singlets, Hadrons, and the Octet of Gluons

At this point representation theory is translated into particle language. That translation is not an afterthought; it is the whole point of the exercise.

A quark field transforms as a 3, an anti-quark field as a $\bar{3}$, and a gluon field as an 8. Of course, these symbols do not mean that particles are literally equal to numbers. They mean only the representation under which the corresponding fields transform.

The gluon deserves special emphasis. With respect to color, it behaves like a quark–anti-quark combination with the singlet removed. That is why the gluon sits in the octet rather than the singlet:

$$3 \otimes \bar{3} = 1 \oplus 8. \tag{4.31}$$

The omitted singlet does not mix with the octet under $SU(3)$, so throwing it away is mathematically consistent.

Now the lecture asks what singlets can be built from these ingredients. The basic answers are:

- a color-singlet qqq state, called a baryon,
- a color-singlet $q\bar{q}$ state, called a meson,
- a color-singlet state built from gluons, called a glueball.

The watchword is color singlets.

Two gluons provide the simplest gluonic example.

The board first records the counting checkpoint

$$8 \times 8 = 1 + 63. \tag{4.32}$$

The meaning is straightforward. Two gluons give 64 states in all; one combination is a singlet, and the remaining 63 states belong to non-singlet sectors.

Later in the discussion the lecture fills this in more explicitly. In cleaned notation one writes

$$8 \otimes 8 = 1 \oplus 8 \oplus 8 \oplus 10 \oplus \overline{10} \oplus 27. \tag{4.33}$$

This should be read cautiously: the screenshot itself shows only the state count, not the full irreducible decomposition. But the later discussion does make clear what the formula means. In an appropriate basis, the 64×64 matrices acting on the two-gluon system become block diagonal, with separate blocks for the singlet, the two octets, the 10, the $\overline{10}$, and the 27. The one-dimensional block is the glueball sector.

Susskind then sharpens the physical point. The observable hadrons are singlets. Baryons are the singlet states in $3 \otimes 3 \otimes 3$. Mesons are the singlet states in $3 \otimes \bar{3}$. Glueballs are the singlet states in products of gluon representations. All of them are colorless.

This is also why the free hadrons carry only integer electric charge. The allowed singlet combinations are built from quark–anti-quark pairs, three-quark combinations, or colorless gluonic composites. Those combinations never leave us with isolated fractional quark charge. In the lecture this is not yet the full dynamical explanation, but it is the first structural payoff of color singlets.

There is also a useful analogy here. A singlet under $SU(3)$ is the color analogue of a neutral object under $U(1)$. In electrodynamics an electron transforms by a phase $e^{i\theta}$, a positron by $e^{-i\theta}$, and the phases cancel in the neutral pair. Likewise, a color singlet is the combination on which the group action cancels out completely.

4.7 Why Only Singlets Survive

Now the lecture turns the screw. Up to this point we have classified states. But classification alone does not tell us why only singlets show up in nature as free particles. Are the other representations merely unlikely? The lecture's answer is more severe: the issue is not probability but energy.

Susskind first phrases this provocatively by saying that the energy of the non-singlet states is effectively infinite. He then explains what that means.

The story begins with Nambu. Well before quantum chromodynamics was fully established, Nambu realized that if physical states were forced to be color singlets, then fractionally charged free particles would never appear. That was the original motivation. The next question was what kind of dynamics could actually enforce such a rule.

The lecture answers by analogy with electrodynamics. A plus charge and a minus charge attract each other, and a neutral system has less field energy than a system with net charge. One way to see this is through field lines: a net charge leaves field stretching out to infinity, and that field carries energy. A neutral pair can cancel its field outside the system and therefore has lower energy. If the electromagnetic coupling were enormously larger than it actually is, isolated electric charges might themselves be effectively confined.

Nambu's idea was to transfer this logic to color. The color generators are the sources of the gluon field, just as electric charge is the source of the electromagnetic field. At the lecture's level of description, a gauge theory is precisely a theory in which conserved quantities source Maxwell-like, photon-like fields. In QCD the relevant conserved quantities are the eight color generators, and the corresponding gauge fields are the eight gluon fields.

But here the analogy with electrodynamics breaks in a crucial and fruitful way. Photons are neutral

and therefore do not source one another. Electrodynamics is linear in that sense. Gluons, by contrast, transform in the octet. They carry color. So the gluon field interacts with itself.

That is the key dynamical fact of the lecture. Because the gluon field is itself colored, its lines of flux do not spread independently the way electric field lines do in a simple dipole pattern. Instead, they attract one another and bundle into a flux tube. The field inside the tube is approximately uniform, which means that the energy per unit length is approximately uniform as well. Hence the energy of a separated quark–anti-quark pair grows roughly linearly with the separation:

$$V(r) \propto r. \tag{4.34}$$

The lecture does not derive a string tension or give a detailed potential. It gives the physical picture, and the picture is enough: color is confining because the flux does not fan out; it gets squeezed into a tube.

This immediately explains why isolated color is not seen. If we try to pull a quark away from an anti-quark inside a meson, the flux tube stretches and the energy rises. Either the quark loses its kinetic energy and is pulled back, or the tube breaks by pair creation:

$$q\bar{q} \longrightarrow (q\bar{q}) + (q\bar{q}). \tag{4.35}$$

In either case we do not get a free quark. What we get is one meson, or two mesons, or a whole cascade of hadrons. When the process repeats after a hard collision, the lecture says, a whole spray of hadrons flies out. That is a jet.

The same physical reasoning leaves room for baryons. Three quarks can combine into a singlet, and the flux lines can come together in threes. The lecture treats that as the dynamical counterpart of the earlier group-theoretic fact that $3 \otimes 3 \otimes 3$ contains a singlet.

At this point Susskind also sharpens the meaning of *gauge theory*. A gauge theory, at lecture level, is one in which conserved quantities are coupled to photon-like fields. The connection to symmetry is immediate: conserved quantities arise from symmetries. He then adds one more physical observation. If flux lines can end only on charges, then the sources are conserved. Otherwise one could destroy a charge and simultaneously alter the field arbitrarily far away, violating the basic causal structure of the theory. So conservation of charge and the behavior of flux lines are not two separate facts; they are two descriptions of the same underlying gauge structure.

This completes the QCD part of the Standard Model as it is developed in the lecture. We now have the eight generators of $SU(3)$, together with the familiar $U(1)$ case of electrodynamics, giving nine conserved quantities and nine gauge fields: the eight gluons and the photon. The remaining ingredient is the extra $SU(2)$ of the Standard Model, together with the Higgs sector, which the lecture postpones to the next meeting. In the lecture's own bookkeeping, we now have most of the working parts of

$$SU(3) \times SU(2) \times U(1), \tag{4.36}$$

but not yet the full package.

4.8 Summary

The lecture starts by reopening group theory, not by leaving it behind. Its first job is to separate the abstract symmetry group from the matrix representation without breaking the relation between

them. From there it reviews the dependence of representation dimension on the system, re-derives the $SU(2)$ conditions of unitarity and determinant one, and then shows that the infinitesimal generators are traceless Hermitian matrices, physically the angular momentum operators.

Only after that spine is secure does the lecture repeat the same pattern for $SU(3)$. Quarks live in the 3, anti-quarks in the $\bar{3}$, and gluons in the 8. The tensor-product decompositions $3 \otimes 3$, $3 \otimes \bar{3}$, $3 \otimes 3 \otimes 3$, and $8 \otimes 8$ then explain how mesons, baryons, and glueballs arise as color singlets. Finally the lecture answers the physical question that the algebra alone cannot answer: why only singlets? Because the gluon field carries color, it self-interacts, squeezes flux into tubes, and makes isolated color energetically inaccessible. That is the group theory of QCD carried over into dynamics, and it is the point from which the Standard Model can now be finished.

Chapter 5

New Revolutions in Particle Physics: Standard Model

This lecture begins by widening the frame. We have already studied quantum electrodynamics and quantum chromodynamics, and both are gauge theories; now the question is what that statement really means. The answer is given in a deliberately economical way. We are not asked for a full theory of gauge symmetry. We are asked instead for the minimal mathematical content of a gauge theory, and the lecture unfolds that content in a very specific order: Maxwell first, then the non-abelian lesson of QCD, then the comparison of scales that makes the weak interactions look mysterious, and only after that the weak $SU(2)$ structure and the W bosons.

5.1 Maxwell Theory As The Minimal Gauge-Theory Prototype

The first example is Maxwell theory, because it is the cleanest gauge theory and the least burdened way to begin. Electromagnetism has six field components, three electric and three magnetic, but those six components may be repackaged in terms of a four-potential,

$$A_\mu = (A_0, \vec{A}). \tag{5.1}$$

The point is not that the four-potential will do all the work later in the lecture. In fact Susskind says at once that it will not play a large formal role that evening. The point is simply that the electromagnetic field can be described either in terms of $(\mathbf{E}, \mathbf{B})$ or in terms of A_μ, and that the latter language is the one which generalizes.

The time component A_0 is introduced first. It is the electrostatic potential, the quantity whose spatial derivatives determine the electric force. The lower line on the board is not fully legible, but the intended content is clear enough that we may record it cautiously as

$$F_x = e\,\partial_x A_0, \tag{5.2}$$

with the understanding that the lecture is emphasizing the derivative structure rather than pausing over sign conventions. The magnetic field comes from derivatives of the spatial vector potential, and more specifically

$$\mathbf{B} = \nabla \times \mathbf{A}. \tag{5.3}$$

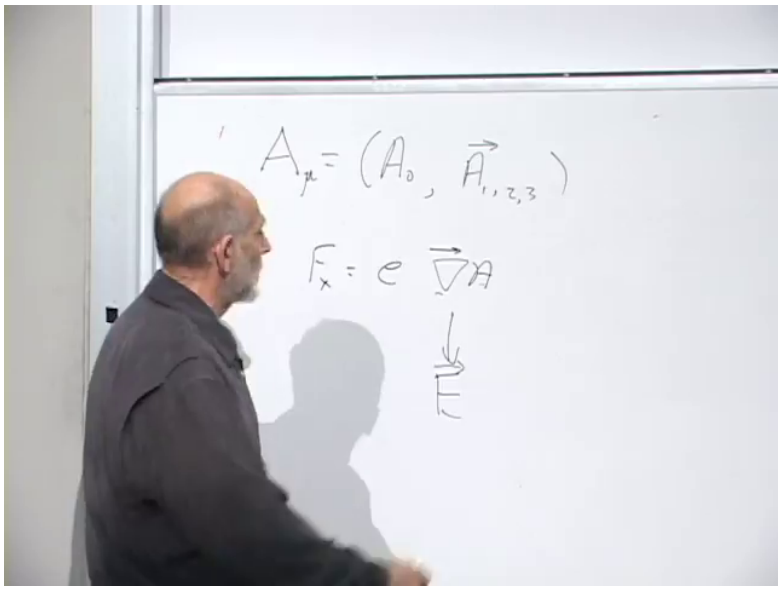


Figure 5.1: Four-vector potential and derivative relation on the board.

Thus there are two equivalent descriptions of electrodynamics: one in terms of six field components, and one in terms of four potentials. The lecture immediately cashes this out dynamically. Maxwell's equations support electromagnetic waves, and a plane-polarized wave has the transverse structure

$$\mathbf{E} \perp \mathbf{B}, \qquad \mathbf{E} \perp \hat{k}, \qquad \mathbf{B} \perp \hat{k}, \tag{5.4}$$

where $\hat{k}$ denotes the direction of propagation. In the normalization Susskind has in mind for the sketch, the electric and magnetic fields have equal magnitude,

$$|\mathbf{E}| = |\mathbf{B}|. \tag{5.5}$$

The polarization is then read from the electric field. This is not presented as a detached textbook reminder. It is the first step in a pattern: identify the field variables, show how they propagate, and then ask what their sources are.

5.2 From Electromagnetism To General Gauge Theory

Once the wave picture is on the board, the lecture turns to the fact that electromagnetic fields are sourced. Charges create electric fields, and electric field lines do not simply end in empty space. They end on charges. If we imagine a sphere surrounding an isolated charge, the flux through that sphere is independent of the sphere's size. In unit-free form, which is the right level for the lecture, we may summarize the point as

$$\oint_{S^2} \mathbf{E} \cdot d\mathbf{S} \propto Q. \tag{5.6}$$

The interesting part is the consequence. If field lines may terminate only on charges, then an isolated charge cannot simply disappear. One cannot make the Coulomb field vanish everywhere at once, since that would require instantaneous causal propagation, and one cannot let the field lines stop without leaving a source behind. The lecture therefore uses Gauss's law not merely as a formula, but as the physical intuition behind charge conservation.

This is the point at which the chapter can say, in the lecture's own spirit, what a gauge theory is. A gauge theory is a theory of Maxwell-like fields: fields with electric-like and magnetic-like components, fields that can propagate as waves when weakly coupled, and fields that come with sources whose conservation is tied to symmetry. If one passes to the quantum description, the wave becomes a stream of quanta. In electrodynamics those quanta are photons, and they come with polarization. One may even think of the Coulomb field, roughly speaking, as arising from the repeated emission and absorption of photons.

The conservation law is then tied to a symmetry of the matter field. For electric charge the symmetry is the familiar phase rotation

$$\psi \to e^{i\theta}\psi. \tag{5.7}$$

The lecture deliberately avoids writing down the Lagrangian, but it later says explicitly what is meant: the Lagrangian would be invariant under the corresponding transformation. For present purposes the shorter language is enough. Conserved source and symmetry belong to the same structure.

The lecture then inserts a useful counterexample. Baryon number is conserved, and one could imagine it acting as a gauge charge. If it did, protons and neutrons would feel a Coulomb-like long-range force associated not with electric charge but with baryon number. Like baryon numbers would repel; opposite baryon numbers would attract. But nature does not supply such a gauge field. The example is mathematically possible and physically false. It clarifies the point that conserved quantities do not automatically come with gauge fields, though genuine gauge charges do.

5.3 QCD As The Non-Abelian Model Case

With that general picture in place, the lecture returns to QCD as the richer non-abelian example. Quarks transform in the fundamental representation of $SU(3)$, and the board writes this in the form

$$q_i' = U_{ij} q_j. \tag{5.8}$$

The index i is the color index: red, green, or blue. Anti-quarks are represented by the complex-conjugate fields, so if we conjugate the transformation law we obtain

$$q_i'^* = U_{ij}^* q_j^*. \tag{5.9}$$

This is precisely the point supported by the second selected frame: the original transformation and the conjugate transformation are written one under the other.

The isolated A_{ij} on the left side of the frame belongs to the neighboring gluon discussion, not to the derivation of the conjugate rule itself. But that neighboring discussion is exactly where the lecture now goes. The QCD gauge field is denoted A_{ij}, a matrix-valued field with one quark-type index and one anti-quark-type index. In transformation language it behaves as though a quark and an anti-quark were sitting on top of one another, not in spin or mass, but in color structure.

A general 3×3 matrix would have nine components, but one removes the singlet piece by imposing tracelessness:

$$\operatorname{tr} A = A_{ii} = 0. \tag{5.10}$$

That is why there are eight gluons rather than nine. In general,

$$\dim(\text{adjoint of } SU(N)) = N^2 - 1. \tag{5.11}$$

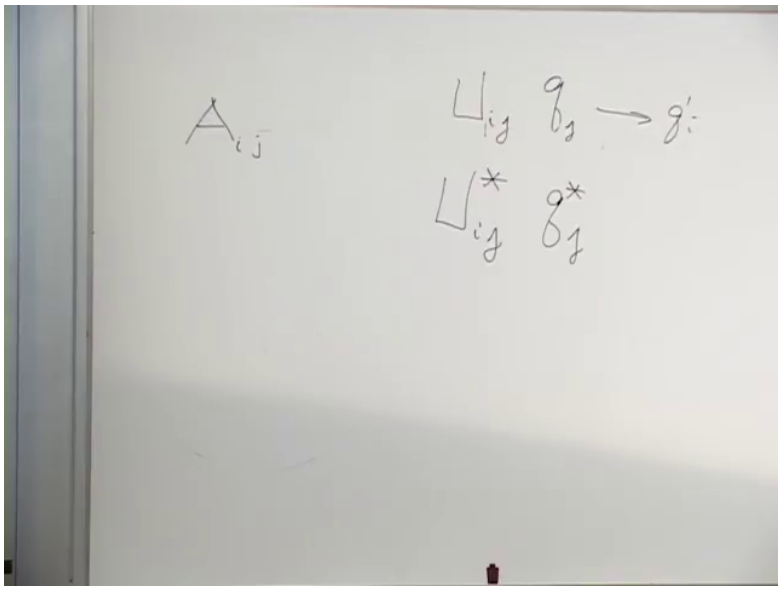


Figure 5.2: Complex-conjugate $SU(3)$ transformation rule.

The lecture then shifts from representation theory to interaction rules. If we follow the indices through a diagram, an i-type quark can become a j-type quark by emitting the corresponding gluon:

$$q_i \to q_j + A_{ij}. \tag{5.12}$$

Susskind momentarily wavers on the order of the indices when speaking, but the real rule is not in doubt: one follows the lines and never loses an index.

Now comes the genuinely new feature. Because the gluon field itself carries a quark-type and an anti-quark-type index, gluons carry color. Unlike photons, they are not neutral under the corresponding gauge symmetry. That means they can interact with each other. The lecture encodes this schematically by following the same index flow through a gluon line, so that one may think in terms of an ij-gluon turning into an ik-gluon together with a kj-gluon. The point is not the exact algebraic form of a Yang–Mills vertex; the lecture explicitly avoids writing that machinery. The point is that gluons exert forces on gluons.

This is then converted into the qualitative flux-tube picture. Field lines still cannot end, but now they attract one another and bundle. The result is confinement: as two quarks are pulled apart, the energy in the tube of color flux grows with distance. QCD is therefore not merely another example of a gauge theory. It is the example that shows how radically non-abelian dynamics can differ from electrodynamics.

5.4 Coupling Constants, Scales, And Why "Strong" Means Strong

At this stage the lecture pauses for a brisk summary of gauge theory: Maxwell-like fields, sources, Gauss's law, conservation, symmetry, and emission or absorption rules. Then it adds one more ingredient, the coupling constant. In electrodynamics the electric charge is the amplitude for photon emission at a vertex. Probability is the square of amplitude, so operationally

$$P(\text{photon emission}) \propto e^2. \tag{5.13}$$

Up to the ubiquitous factors of 4π, one packages this into the fine-structure constant,

$$\alpha \sim \frac{e^2}{4\pi} \approx \frac{1}{137}. \tag{5.14}$$

This is the lecture's concrete interpretation of charge: not merely a label, but a measure of how likely emission is.

The next movement may look like a digression, but it is really the preparation for the weak force. We are reminded that in natural units

$$c = 1, \qquad \hbar = 1, \qquad E \sim \omega \sim \frac{1}{t}, \qquad x \sim \frac{1}{E}. \tag{5.15}$$

Distance, time, and inverse energy are then the same kind of quantity measured in different languages.

With that in hand, the lecture compares atoms and hadrons. For atoms the relevant scales are of order

$$R_{\text{atom}} \sim 10^{-10}\,\text{m}, \tag{5.16}$$
$$t_{\text{transit}} \sim 10^{-18}\,\text{s}, \tag{5.17}$$
$$t_{\text{orbital}} \sim 10^{-16}\,\text{s}, \tag{5.18}$$
$$t_{\text{decay}} \sim 10^{-9}\,\text{s}. \tag{5.19}$$

The striking point is not the individual numbers but their hierarchy. The electron moves slowly compared with light, because the electromagnetic force is weak, and once accelerated it still emits photons inefficiently because α is small. The decay time is therefore suppressed twice over: once by the small acceleration, once by the small emission probability.

For hadrons, by contrast, the scales collapse toward one another:

$$R_{\text{hadron}} \sim 10^{-15}\,\text{m}, \tag{5.20}$$
$$t_{\text{transit}} \sim 10^{-23}\,\text{s}, \tag{5.21}$$
$$t_{\text{orbital}} \sim 10^{-23}\,\text{s}, \tag{5.22}$$
$$t_{\text{decay}} \sim 10^{-23}\,\text{s}. \tag{5.23}$$

There is no comparable hierarchy. The lecture's conclusion is therefore dynamical rather than taxonomic:

$$\alpha_{\text{QCD}} \gg \alpha. \tag{5.24}$$

At lecture level one may think of α_{QCD} as lying around 0.2, or more loosely as being much closer to one than 1/137. That is why the strong interaction is called strong. It proceeds about as fast as such a process can proceed.

5.5 Why Weak Interactions Are Puzzling Before We Even Build Them

Only after the gauge-theory recap and the comparison of scales do we reach the weak interaction. This order matters. The lecture wants the reader to feel the puzzle before seeing the mechanism.

Historically weak processes appeared through beta decay, and the old name still captures the phenomenon well enough: the decay rates are extraordinarily slow.

The first example is neutron beta decay,

$$n \to p + e^- + \bar{\nu}_e, \qquad \tau_n \sim 12 \text{ minutes}. \tag{5.25}$$

On hadronic timescales that is absurd. Part of the explanation is obvious. The neutron lies only slightly above the proton-plus-electron threshold, with the lecture quoting the rough numbers

$$m_n \approx 940\,\text{MeV}, \qquad m_p \approx 939\,\text{MeV}, \qquad m_e \approx 0.5\,\text{MeV}. \tag{5.26}$$

So the available energy is tiny. If the neutron were just a little lighter, it could not decay at all. If it were only infinitesimally above threshold, the decay would indeed be slow. But the lecture insists that this is only a small part of the story. Twelve minutes is far too long to be explained by phase space alone.

The second example is a charged pion. The lecture first uses the channel

$$\pi^- \to e^- + \bar{\nu}_e, \qquad \tau_{\pi^-} \sim 10 \text{ ns}, \tag{5.27}$$

not because that is the dominant decay, but because it is a simple first route into the weak vertex. Here the kinematic situation is entirely different. There is plenty of energy available. So the slowness cannot be blamed on near-threshold suppression. Something else must be making weak processes proceed reluctantly.

That is the real setup for the second half of the lecture. The question is not merely what particles are involved, but what sort of symmetry could produce these decays at all.

5.6 Horizontal Doublets, Leptons, And The Weak $SU(2)$

To answer that, the lecture builds a table. The transcript is garbled at the moment when the table first appears, but the stable structure is clear from the surrounding explanation. Quarks come in weak doublets,

$$\begin{pmatrix} u \\ d \end{pmatrix}, \qquad \begin{pmatrix} c \\ s \end{pmatrix}, \qquad \begin{pmatrix} t \\ b \end{pmatrix}, \tag{5.28}$$

and leptons do as well,

$$\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}, \qquad \begin{pmatrix} \nu_\mu \\ \mu^- \end{pmatrix}, \qquad \begin{pmatrix} \nu_\tau \\ \tau^- \end{pmatrix}. \tag{5.29}$$

On the board this was arranged so that color transformations could be pictured as acting vertically through the quark entries, while the weak symmetry acted horizontally, mixing the members of each paired column. In cleaned notation it is simpler to display the pairs as doublets, but the lecture's geometric point should be remembered: color and weak symmetry act in different directions.

The charged members of the quark doublets have charges $+2/3$ and $-1/3$, while the lepton doublets have charges 0 and -1. In both cases the change across the doublet is one unit:

$$Q(u) - Q(d) = +1, \qquad Q(\nu_e) - Q(e^-) = +1. \tag{5.30}$$

That common charge step is one of the reasons the quark and lepton sectors can be spoken of in a single breath here.

The natural group is then $SU(2)$, because the symmetry acts on doublets. The lecture does not yet want the formal representation theory of $SU(2)$ written out in detail. What matters immediately is that an $SU(2)$ gauge theory should have three generators, hence three gauge bosons. At this stage only $W^{\pm}$ are built explicitly. The third boson is acknowledged as lurking in the background and postponed.

5.6.1 Question & Answer

Question. Charge conservation tells us that the boson in processes such as $e^- \to \nu_e + \cdots$ and $d \to u + \cdots$ must carry charge -1. But why should it be the same boson in all of these cases?

Answer. The lecture's answer is that this is an empirical fact encoded by the symmetry itself. The weak interaction does not act by one $SU(2)$ on electrons, another on quarks, and a third on the heavier families. It acts simultaneously on all the doublets. Nature could have been more complicated, with several different W-type bosons attached to different rows, but experimentally it is simpler: the same gauge bosons mediate all these horizontal transitions.

5.7 $W^{\pm}$ Bosons, Decay Diagrams, And Off-Shell Intermediate States

Now the lecture builds the charged gauge bosons operationally. One looks for an object with the quantum numbers of a particle from the second column together with the antiparticle of a particle from the first. On the lepton side that means $e^-\bar{\nu}_e$. On the quark side it means $d\bar{u}$. Either way the electric charge is -1, so the object is called W^-:

$$W^- \sim e^-\bar{\nu}_e \sim d\bar{u}. \tag{5.31}$$

Again, the symbol $\sim$ is only a statement about quantum numbers, not a claim that the W is literally a bound state of those particles. Likewise

$$W^+ \sim e^+\nu_e \sim \bar{d}u. \tag{5.32}$$

The basic weak vertices are then

$$e^- \to \nu_e + W^-, \tag{5.33}$$
$$d \to u + W^-, \tag{5.34}$$
$$s \to c + W^-. \tag{5.35}$$

Once one vertex has been written, many other processes follow by flipping lines and arrows. The lecture is careful about this point: when a line is reversed in time, particle and antiparticle are exchanged. That is why the same local interaction can be read as emission, absorption, decay, or inverse decay, depending on how the diagram is oriented.

The first worked example is the pion. Since

$$\pi^- \sim d\bar{u}, \tag{5.36}$$

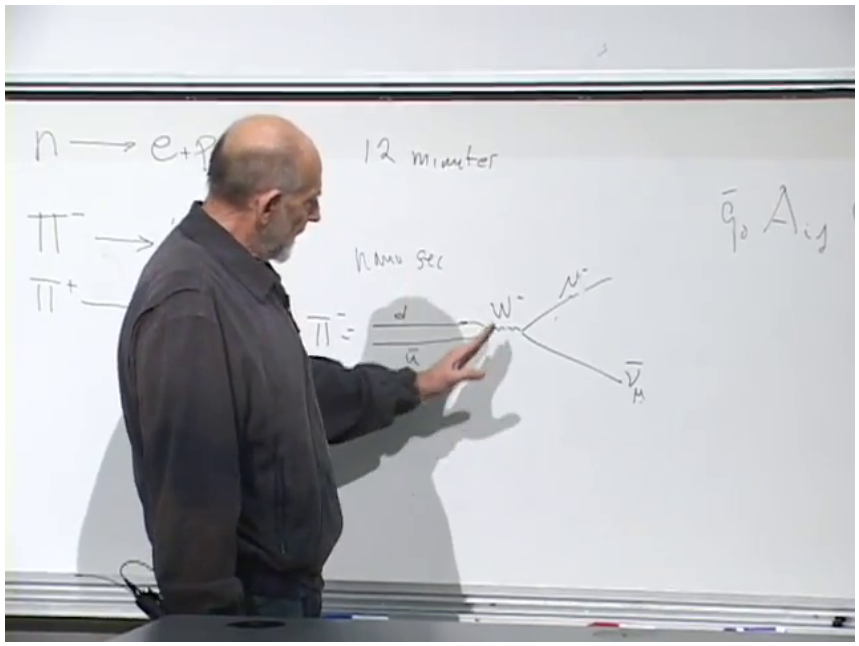


Figure 5.3: Pion decay through an intermediate W^-.

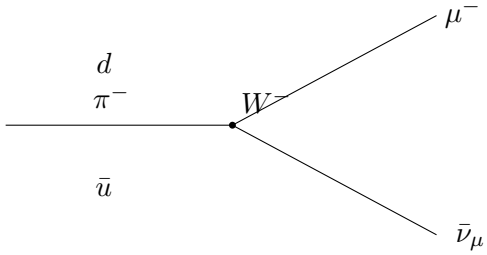


Figure 5.4: Clean redraw of the decay topology used in the lecture's kinematic discussion.

the pion can turn into a W^-, and the W^- can then decay. The lecture first draws the electron channel because it exposes the weak vertex cleanly:

$$\pi^- \to W^- \to e^- + \bar{\nu}_e. \tag{5.37}$$

Only after that does it add the phenomenological correction: the dominant charged-pion decay is actually

$$\pi^- \to W^- \to \mu^- + \bar{\nu}_\mu. \tag{5.38}$$

The lecture even pauses to note that the diagram may be read in reverse. A pion created by strong interactions can decay into a muon and antineutrino, but if one were to fire a muon and an antineutrino toward one another with the right kinematics, they could also produce a π^-. That is exactly the sort of two-way reading one expects from a Feynman diagram.

The next example is the muon, and here the process is purely leptonic:

$$\mu^- \to \nu_\mu + W^- \to \nu_\mu + e^- + \bar{\nu}_e. \tag{5.39}$$

The muon is unstable simply because it is heavy enough to decay to the electron. The electron is

stable against that mode because there is nothing lighter with the same charge pattern for it to decay into.

Then comes neutron beta decay at quark level. Writing the neutron as ddu, one possibility would be to turn the up quark into a down quark. But that would create a three-down-quark state, and the lecture rejects that route because it is too heavy. The successful route is

$$ddu \to uud + W^-, \tag{5.40}$$

$$uud = p, \tag{5.41}$$

$$W^- \to e^- + \bar{\nu}_e. \tag{5.42}$$

Therefore

$$n \to p + e^- + \bar{\nu}_e. \tag{5.43}$$

In the lecture's own words, the basic components are always the same: one column-changing transition, followed by a W, followed by another column-changing transition.

5.7.1 Question & Answer

Question. How can the intermediate W fail to conserve energy?

Answer. The lecture answers this at the level of Feynman-diagram kinematics. One does not insist that every intermediate line be on-shell. Exact conservation is enforced between the initial and final states. An intermediate state may violate the on-shell relation for a short time, and the lecture summarizes that by the energy-time uncertainty principle,

$$\Delta E \, \Delta t \sim 1. \tag{5.44}$$

So one should not demand exact energy conservation between the first vertex and the second. One demands it for the complete process.

The pion example makes this concrete. If the pion is at rest, then

$$m_\pi \approx 140\,\text{MeV}, \qquad m_\mu \approx 100\,\text{MeV}, \qquad m_\nu \approx 0. \tag{5.45}$$

Hence the final rest masses do not exhaust the initial energy,

$$m_\pi > m_\mu + m_{\nu_\mu}, \tag{5.46}$$

and the difference appears as kinetic energy of the outgoing particles. If the initial pion is at rest, momentum conservation then forces the muon and antineutrino to emerge back-to-back.

Exactly the same logic explains why ordinary neutron beta decay does not produce a muon. There is not enough available energy. If one were to excite the neutron strongly enough, a muon channel could open, but for an unexcited neutron only the electron channel is kinematically accessible.

5.8 Summary

The lecture has now unfolded its main construction. A gauge theory means Maxwell-like fields, sources, conservation laws, symmetries, and the emission or absorption rules that go with them.

Maxwell theory supplied the prototype, QCD supplied the non-abelian example in which gauge bosons themselves carry charge and interact, and the weak interaction supplied a new $SU(2)$ structure acting on doublets of quarks and leptons.

But the final note is deliberately unresolved. One might guess that weak decays are slow because the weak analog of the fine-structure constant is ridiculously small. The lecture says explicitly that this is not the reason. At the level relevant here, the weak coupling is not absurdly smaller than the electromagnetic one. So the local machinery of the weak interaction is now on the board, but the real mystery is still ahead: why are weak decay rates so slow? That puzzle is left standing for the next lecture.

Chapter 6

New Revolutions in Particle Physics: Standard Model

We begin exactly where the lecture resumes: with $SU(3)$ and $SU(2)$, with the fermion table already on the board, and with the insistence that a gauge symmetry should be understood through what it does. These notes follow Leonard Susskind's lecture, curated by LazyingArt LLC, and preserve its central line of thought. Weak interactions are weak, but not because the weak coupling is fantastically tiny. The suppression comes from the propagation of a very heavy W boson. Only after that argument has been made does the lecture turn, qualitatively and deliberately unfinished, toward spontaneous symmetry breaking.

6.1 Gauge Symmetries on the Fermion Table

The lecture does not begin from representation theory. It begins from a table. Along the horizontal direction sit the quark families, grouped as weak doublets,

$$(u, d), \qquad (c, s), \qquad (t, b),$$

and along the vertical direction sit the three colors,

$$r, \qquad g, \qquad b.$$

This is not the table of all elementary particles; it is the table of the Standard Model fermions that make up atoms and nuclei, organized so that the symmetries can be seen acting on them.

What matters first is the direction of the action:

- $SU(3)$ acts vertically, mixing red, green, and blue while leaving the family label alone.
- $SU(2)$ acts horizontally within each doublet, moving $u \leftrightarrow d$, $c \leftrightarrow s$, and $t \leftrightarrow b$, while leaving color alone.
- $U(1)$ acts by phase on every charged field.

So, at this stage, the Standard Model is presented as the product gauge structure

$$SU(3) \times SU(2) \times U(1). \tag{6.1}$$

The word "product" is not a formality. It means that the color symmetry does not mix upness and downness, and the weak symmetry does not mix color.

Susskind then makes a stronger point. The generators of a gauge symmetry are not merely algebraic objects. The gauge bosons are their physical realization. When a quark emits a gluon and changes color, the gluon has enacted one of the infinitesimal transformations of the color group. If a red up quark becomes a green up quark, that is one generator at work; and if the symmetry is truly a symmetry of the theory, the same generator must act on every row. The same gluon that changes red up into green up also changes red top into green top. The color move is vertical and it is universal across the table.

The lecture then pivots by contrast. Weak bosons do something gluons do not do:

$$u \to d + W^+, \qquad d \to u + W^-. \tag{6.2}$$

A gluon never moves us horizontally. A W boson never moves us vertically. That directional contrast is the way the slogan $SU(3) \times SU(2) \times U(1)$ becomes concrete.

6.2 From Quarks to Weak Processes

Once the horizontal action has been identified, the lecture asks how general it is. It is not confined to quarks. The leptons repeat the same family pattern,

$$(\nu_e, e^-), \qquad (\nu_\mu, \mu^-), \qquad (\nu_\tau, \tau^-),$$

but they carry no color and therefore do not participate in the $SU(3)$ interaction. Weakly, however, they fit the same horizontal story.

At this point the lecture emphasizes a specific criterion: W emission tracks the difference of charge across a doublet. For quarks,

$$Q(u) - Q(d) = \frac{2}{3} - \left(-\frac{1}{3}\right) = 1, \tag{6.3}$$

and for the electron-neutrino doublet,

$$Q(\nu_e) - Q(e^-) = 0 - (-1) = 1. \tag{6.4}$$

That is why the same W structure can connect quark pairs and lepton pairs. The weak interaction, as presented here, cares about the correct charge difference, not about whether the fermions carry color.

To make the discussion concrete, the lecture settles on neutron beta decay. At the elementary level,

$$d \to u + W^-, \qquad W^- \to e^- + \bar{\nu}_e, \tag{6.5}$$

so that, at the hadronic level,

$$n \to p + e^- + \bar{\nu}_e. \tag{6.6}$$

Inside the neutron, one down quark turns into an up quark, while the remaining quarks simply carry the neutron into a proton.

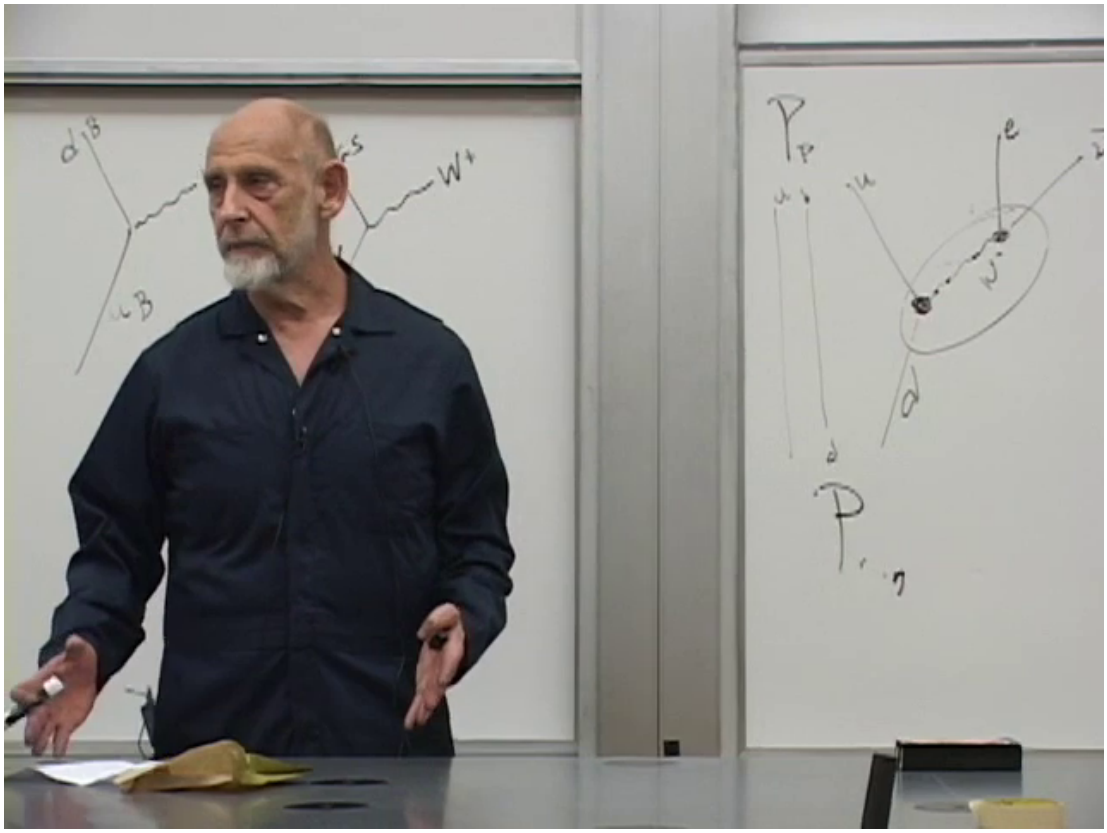


Figure 6.1: Weak-interaction board sketches across both panels. The screenshot is kept as documentary evidence of the lecture's board layout; the labels are only partly legible, but the process topology is clear.

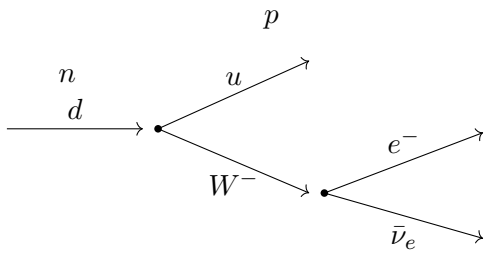


Figure 6.2: A cautious redraw of the weak process underlying neutron beta decay. The spectator-quark bookkeeping is suppressed so that the elementary weak step remains in the foreground.

The board image should stay in view because it preserves the lecture's geometry. But the clean mathematical core is the topology of the weak process, which we can redraw without pretending that every blackboard label is completely secure.

Now the lecture asks a different sort of question. It is no longer asking what processes are allowed. It is asking how probable they are. Electromagnetically, the relevant small quantity is set by the electric charge:

$$e^2 \sim \alpha. \tag{6.7}$$

For gluons the coupling is much larger, so gluon emission is correspondingly much more likely. But the striking point is that the weak coupling is not parametrically tiny compared with the electromagnetic one. In the rough order-of-magnitude sense of the lecture,

$$g_w^2 \text{ is of the same general order as } e^2. \tag{6.8}$$

6.2.1 Question & Answer

Question. If g_w is not especially small, why are weak interactions weak?

Answer. Because the vertex is only half the diagram. A weak process is not suppressed mainly at the point of emission. It is suppressed by what happens in between emission and absorption. To see that, we have to stop looking only at couplings and start looking at propagators.

6.3 Propagators in Distance Space and Natural Units

The lecture now changes mathematical gear. A Feynman diagram has vertices, but it also has propagators. In the present example, the crucial object is the amplitude for the W to go from the point where it is emitted to the point where it turns into the electron and antineutrino. Susskind denotes the position-space propagator by Δ and treats it first as a function of the separation between those two events.

At short distance the propagator is large. Schematically, the lecture describes a massless-style behavior of the form

$$\Delta(d) \propto \frac{1}{d^2} \qquad \text{for small } d. \tag{6.9}$$

If the exchanged particle had no mass, the power law would simply continue. But a massive particle behaves differently. The mass introduces a new ingredient, and after some characteristic distance the propagator turns over and falls exponentially fast. The lecture does not derive an exact closed form here; it only wants the qualitative structure:

$$\Delta(d) \text{ has a power-law regime at short distance and an exponentially suppressed regime at large distance.} \tag{6.10}$$

This is where natural units enter, and in this lecture they are not a side remark. They are the bridge to the distance scale. The blackboard writes

$$c = \hbar = 1, \qquad [m] = [E] = [p] = [1/\text{length}]. \tag{6.11}$$

That is exactly the combination one sees on the board, with $[p]$ written separately and naturally completed into the standard units chain.

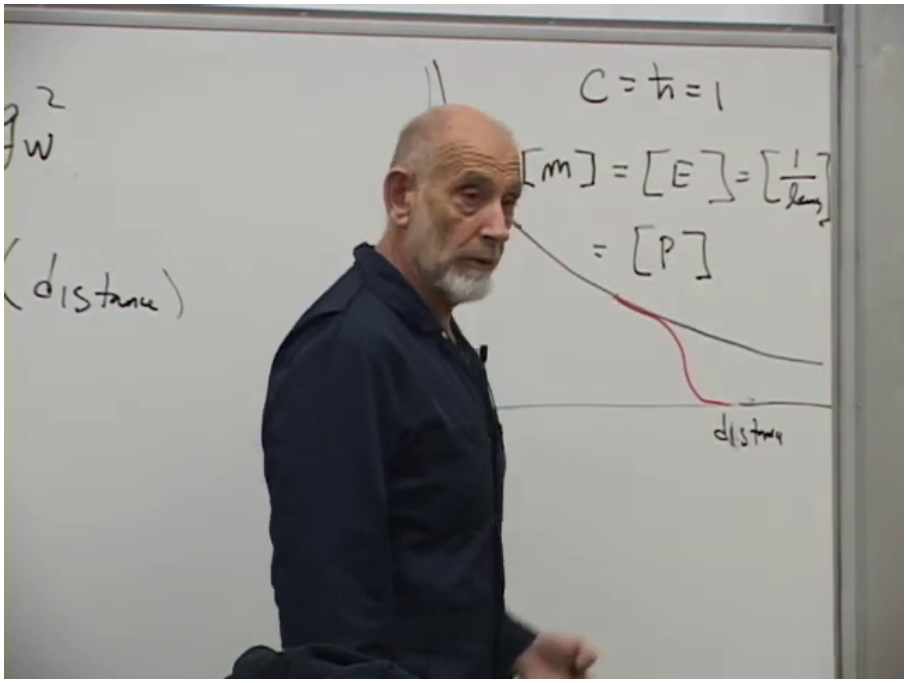


Figure 6.3: Natural-units dimensional relations and the qualitative distance-space sketch of propagator falloff. The screenshot is retained because the lecture ties the units statement directly to the graph.

Once the units are in place, the rest is dimensional analysis. If length has the dimensions of inverse mass, then the only distance we can build from a mass m is

$$d \sim \frac{1}{m}. \tag{6.12}$$

With constants restored, this becomes

$$d \sim \frac{\hbar}{mc}, \tag{6.13}$$

the Compton wavelength scale. That is the distance at which the mass of the exchanged particle begins to matter and the propagator dives.

In the lecture's own language, the vertical axis of the sketch is "the size of Δ," and the square of the propagator is then used as a measure of the probability that the particle can get that far before converting. One should not overread that statement as a precise general QFT formula, but it captures exactly the intended lesson: the heavier the exchanged particle, the shorter the distance over which its propagation remains effective.

We now know, qualitatively, why a heavy exchanged particle makes a long-distance force ineffective. The lecture's next move is to translate that picture into the momentum variables that experiments actually measure.

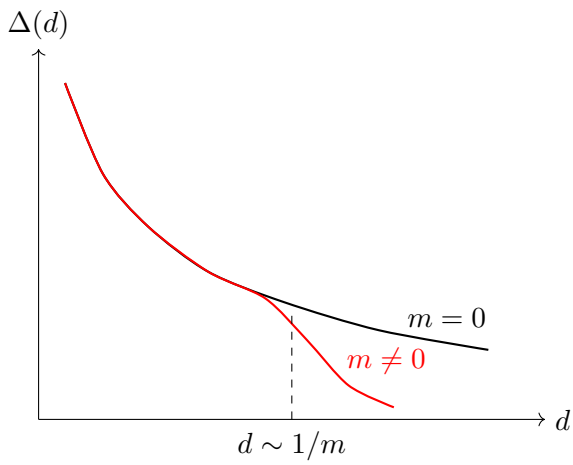


Figure 6.4: A schematic redraw of the lecture's distance-space picture: one slow power-law tail, and one curve that turns into much sharper suppression once the mass scale is felt.

6.4 Momentum-Space Propagator and the Low-Energy Weak Estimate

What is prescribed in an experiment is generally not the separation between the emission and absorption points. It is the momenta of the external particles. So the lecture passes from the position-space propagator to its Fourier transform. Susskind explicitly says that the twiddle indicates Fourier transform, and he lets k denote the momentum carried by the exchanged W.

In the lecture's simplified low-energy convention,

$$\widetilde{\Delta}_W(k) = \frac{1}{k^2 + m_W^2}. \tag{6.14}$$

For a photon, which has no mass, this becomes

$$\widetilde{\Delta}_\gamma(k) = \frac{1}{k^2}. \tag{6.15}$$

The lecture briefly acknowledges the fuller sign convention of relativistic propagators, but the beta-decay kinematics are mostly space-like and, more importantly, the transferred momentum is minuscule compared with m_W. So the denominator is dominated by the mass term regardless.

Remark 6.1. We keep the lecture's low-energy form $\widetilde{\Delta}_W(k) = 1/(k^2 + m_W^2)$ rather than silently switching to a full Minkowski-signature treatment. The point here is the low-momentum suppression by m_W, not a general propagator tutorial.

Now the weak-rate estimate falls into place. The diagram for beta decay contains two weak vertices and one W-propagator. Therefore

$$\mathcal{A}_{n\to pe^-\bar{\nu}_e} \sim g_w\, \widetilde{\Delta}_W(k)\, g_w \tag{6.16}$$

$$\sim \frac{g_w^2}{k^2 + m_W^2}. \tag{6.17}$$

But in beta decay,

$$k^2 \ll m_W^2, \tag{6.18}$$

so the propagator can be replaced, to excellent approximation, by its low-momentum limit:

$$\widetilde{\Delta}_W(k) \approx \frac{1}{m_W^2}. \tag{6.19}$$

Hence

$$\mathcal{A}_{n\to pe^-\bar{\nu}_e} \sim \frac{g_w^2}{m_W^2}. \tag{6.20}$$

This is the clean low-energy scaling argument of the lecture. It is not yet a fully normalized electroweak formula, and it should not be presented as one.

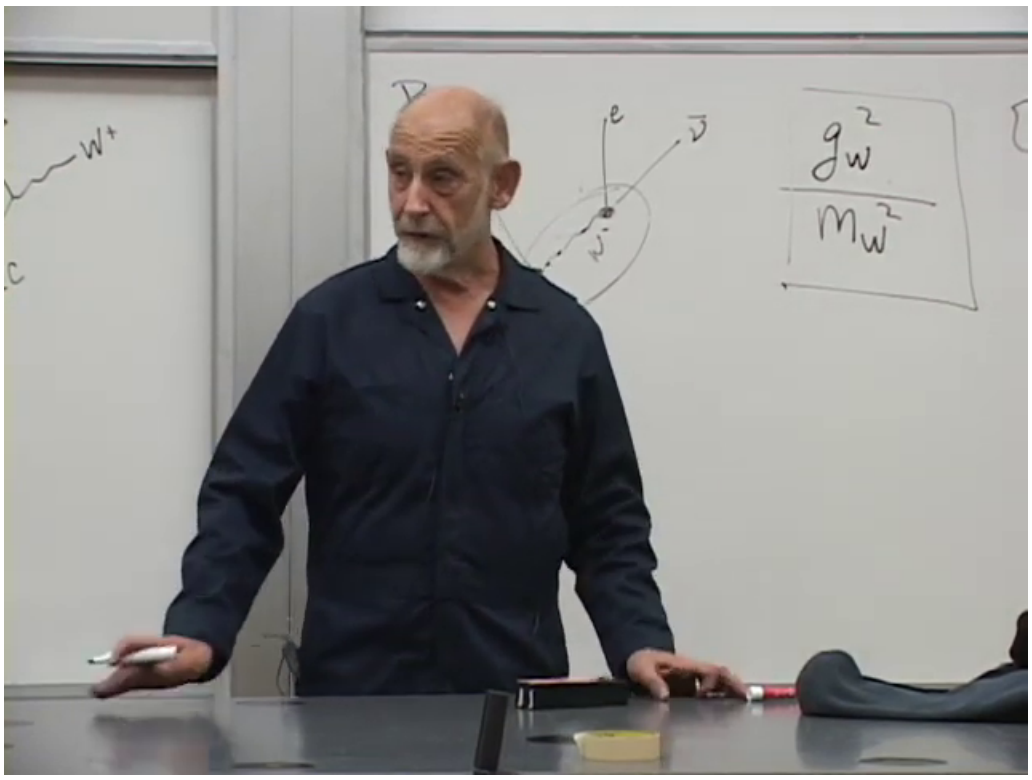


Figure 6.5: Weak-reaction estimate on the board. The boxed ratio is written beside, not inside, the process sketch, exactly as the lecture turns from diagram to scaling estimate.

We may now typeset the cleaned version of the boxed board relation:

$$\frac{g_w^2}{m_W^2}. \tag{6.21}$$

The board letterform oscillates between lowercase-looking and uppercase-looking w/W, but the notes standardize to W and m_W.

Squaring the amplitude gives the scale of the probability:

$$P_{\text{weak}} \sim |\mathcal{A}|^2 \sim \frac{g_w^4}{m_W^4}. \tag{6.22}$$

That is the mathematical payoff of the whole middle part of the lecture. Weak interactions are weak because the W is heavy, not because the weak coupling is fantastically smaller than the electromagnetic one.

There is also an important historical point. Old weak-interaction experiments did not have enough momentum transfer to make the denominator visibly vary with k. What was actually measured was the low-energy combination

$$G_F \sim \frac{g_w^2}{m_W^2}, \tag{6.23}$$

the Fermi constant. Only later, once experiments could reach large enough momenta, did the effective constant reveal its hidden k-dependence.

6.5 Virtual Exchange and Energy Conservation

After the formula has been written down, the lecture slows down. That is exactly the right place to slow down, because the central conceptual objection has now become sharp. If the W is so heavy, how can it appear at all in neutron decay?

6.5.1 Question & Answer

Question. How can a heavy W appear when there is not enough energy to create a real W, and what makes it virtual rather than real?

Answer. It cannot appear as a real outgoing particle in ordinary beta decay. It appears only as an internal line in the amplitude, and then only for a very short time. The lecture encodes this with the energy-time uncertainty estimate

$$\Delta E \, \Delta t \lesssim \hbar. \tag{6.24}$$

If the intermediate state carries an apparent excess energy ΔE, that mismatch can survive only for a time of order

$$\Delta t \lesssim \frac{\hbar}{\Delta E}. \tag{6.25}$$

So the W in beta decay is not a free on-shell particle wandering around the laboratory. It is a short-lived virtual intermediate state.

The lecture makes this vivid with a tunneling analogy. Consider a particle initially at rest in a potential well. The far side of the barrier is chosen to have the same potential energy as the starting point. Then the particle may emerge on the far side and still be at rest, so the initial and final energies match. But inside the barrier the potential energy is higher. If we insist on classical language there, the intermediate stage seems to violate energy conservation. Quantum mechanically, that apparent violation is permitted only briefly.

The transfer back to beta decay is immediate. There is simply not enough energy in a neutron to emit a real W and let it fly off. What can happen is a brief virtual emission. The W then has to disappear quickly: it may be reabsorbed, or it may turn into lighter final particles such as $e^- + \bar{\nu}_e$. In the lecture's language, the heavy W under the propagator is like the particle under the barrier.

Just as important is what *virtual* means. In the lecture's own distinction, real particles are the incoming and outgoing asymptotic states of the process. Virtual particles are the internal lines of the Feynman diagram. They are not directly observed. If one tries to observe them, one must inject enough energy and enough time resolution to change the problem. A probe sharp enough to

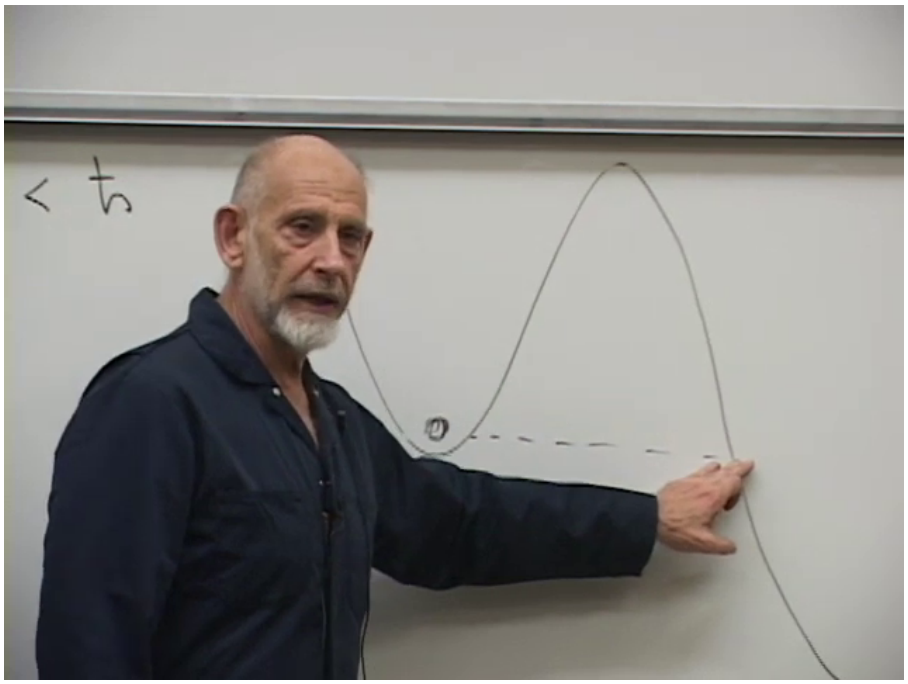


Figure 6.6: Potential barrier and equal-energy tunneling points. The board image matters here because it preserves the exact point on the far slope that the lecture identifies as having the same potential energy as the starting point.

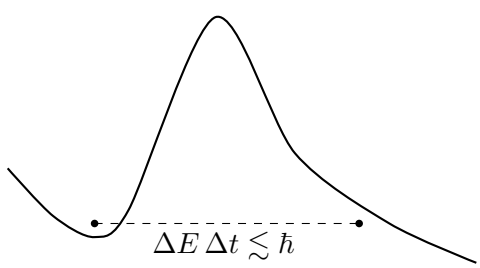


Figure 6.7: A schematic redraw of the tunneling analogy: the marked points have the same total energy, while the barrier region represents the classically forbidden intermediate stage.

resolve the short time Δt also carries enough energy to create the corresponding real particle. One no longer has the same process; one has turned the virtual line into a final, real W.

This is why the lecture insists that exact energy conservation is never abandoned. Between the full initial state and the full final state, conservation is exact. The apparent violation belongs only to the intermediate description.

The audience discussion then sharpens one more point. A real W is not governed by the uncertainty time $\hbar/\Delta E$ of the virtual process. It has its own short proper lifetime, and if it is produced with large momentum it exhibits ordinary time dilation. Weak lifetimes are long only when the process is forced to proceed through a heavy virtual W at low energy.

At this stage the lecture also summarizes rate suppression in a useful way. Three ingredients matter:

- coupling constants,
- propagators, hence masses and momentum dependence,
- available energy.

In ordinary neutron beta decay, the central suppression is the second of these.

6.6 The Missing Third Weak Boson

Once the discussion of weak rates and virtual exchange has been completed, the lecture briefly returns to the symmetry structure. This is not an afterthought. It is a reminder that the group theory has not yet been fully closed.

If the weak interaction is governed by $SU(2)$, then the number of weak gauge bosons must match the number of generators. But

$$\dim SU(2) = 3. \tag{6.26}$$

So W^+ and W^- cannot be the whole story. Two charged bosons are not enough. There must be a third weak gauge boson, neutral rather than charged. The lecture does not work it out here; it points forward instead, saying that the missing boson is closely related to the Z, and that the precise relation among $SU(2)$, $U(1)$, electric charge, and the neutral weak boson will be taken up next time.

This short return matters because it prevents the opening table picture from hardening too early into a complete theory. The lecture keeps reminding us that the real electroweak structure is a little subtler than the first pass suggests.

6.7 Spontaneous Symmetry Breaking

Only late in the lecture does the subject change, and even then the change is really a bridge to the next lecture. We are not given the Higgs mechanism. We are given the minimal qualitative structure that will later make the Higgs mechanism intelligible.

The first illustration is the dinner-table story. A perfectly symmetric arrangement admits two equally natural choices, left and right. Nothing in the local setup prefers one over the other. But the arrangement is unstable, and once one choice is initiated, the whole system settles that way. The illustration is intentionally informal, but it prepares the right intuition.

Then the lecture gives a more mathematical toy model. Take a lattice of two-state objects; they may be spins, but for the moment they may just as well be classical coins showing heads or tails. Neighboring variables prefer to align, so that

$$E_{\text{parallel}} < E_{\text{anti-parallel}}. \tag{6.27}$$

The lowest-energy configurations are therefore the uniformly aligned ones. But there are two of them:

$$E(\text{all heads}) = E(\text{all tails}). \tag{6.28}$$

So the system has two degenerate ground states.

Now comes the characteristic move. If far away one of the coins is fixed to be heads, that tiny distant asymmetry can select the all-heads ground state. Once that choice has been made, local excitations are measured relative to it. In particular, a tail inserted into a sea of heads costs energy:

$$E(\text{one tail in a sea of heads}) > E(\text{all heads}). \tag{6.29}$$

The microscopic rules were symmetric, but the state that the system chose was not. That is the basic phenomenon of spontaneous symmetry breaking.

6.7.1 Question & Answer

Question. How is spontaneous symmetry breaking different from explicit symmetry breaking?

Answer. In explicit symmetry breaking the local rules are already biased. If a single head simply had lower energy than a single tail, then the symmetry would be broken in the Hamiltonian itself, and there would be only one preferred vacuum. In spontaneous symmetry breaking the local rules remain symmetric, the vacua remain degenerate, and it is the choice of ground state that breaks the symmetry.

The lecture then gives the clean diagnostic. Suppose one insists that very far away on one side the system is heads, and very far away on the other side it is tails. If the breaking is spontaneous, the minimum-energy configuration contains an interface between the two regions. That interface is a domain wall. Domain walls are therefore the qualitative signal of spontaneous, rather than explicit, symmetry breaking.

This is where the lecture stops, and it is wise to let it stop there. The ending is intentionally qualitative. It leaves us with a symmetric local theory, a degenerate family of vacua, an instability that selects one of them, and the preview that the Higgs phenomenon belongs precisely to this class of ideas.

6.8 Summary

The lecture opens by rebuilding the Standard Model from the fermion table outward. $SU(3)$ acts vertically on color, $SU(2)$ acts horizontally on weak doublets, and $U(1)$ acts by phase. The weak interaction is then made concrete through neutron beta decay and through the observation that the same W structure acts across quark and lepton families when the charge difference is right.

The central question is then posed in the right form: if g_w is not especially small, why are weak interactions weak? The answer comes from the propagator of a massive exchanged particle. In the lecture's low-energy convention,

$$\widetilde{\Delta}_W(k) = \frac{1}{k^2 + m_W^2},$$

so at low momentum

$$\mathcal{A}_{\text{weak}} \sim \frac{g_w^2}{m_W^2}, \qquad P_{\text{weak}} \sim \frac{g_w^4}{m_W^4}.$$

That is the spine of the lecture.

After the estimate, the lecture turns interpretive again. A heavy W can appear only virtually, for a short time, and the tunneling analogy makes that statement intelligible while exact initial-to-final energy conservation remains untouched. Only then does Susskind return briefly to the unfinished electroweak group structure, noticing that $SU(2)$ should have a third gauge boson. The final turn is qualitative: spontaneous symmetry breaking, degenerate vacua, and domain walls as the signal that the symmetry was broken by the choice of ground state rather than by an explicit local bias. The next lecture will pick up both loose threads: the Z boson and the Higgs phenomenon.